CSA Group

CAN/CSA-O15-15
National Standard of Canada
(approved February 2016)

Wood utility poles and reinforcing stubs



Standards Council of Canada
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REVISED JUNE 2016

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Revision History

O15-15, Wood utility poles and reinforcing stubs

Update No. 1 — June 2016	**Revision symbol (in margin)**
Clause 4.4.12 Figure 1	①

Errata — May 2015	**Revision symbol (in margin)**
Clauses A.8.2 and E.1 Tables 3, A.1, E.3, and E.13	Δ

National Standard of Canada — February 2016
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Standards Update Service

CAN/CSA-O15-15
January 2015

Title: *Wood utility poles and reinforcing stubs*

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Wood utility poles and reinforcing stubs

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Published in January 2015 by CSA Group
A not-for-profit private sector organization
178 Rexdale Boulevard, Toronto, Ontario, Canada M9W 1R3

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ICS 79.080
ISBN 978-1-77139-837-4

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Preface

This is the fourth edition of CSA O15, *Wood utility poles and reinforcing stubs*. It supersedes the previous editions published in 2005, 1990, and 1983.

Major changes in this edition include the method of measurement of sweep in poles and the reference to the ANSI photographic manual.

This Standard was prepared by the Subcommittee on Wood Poles, under the jurisdiction of the Technical Committee on Solid and Engineered Wood Products and the Strategic Steering Committee on Forest Products, and has been formally approved by the Technical Committee.

This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Notes:

1) *Use of the singular does not exclude the plural (and vice versa) when the sense allows.*

2) *Although the intended primary application of this Standard is stated in its Scope, it is important to note that it remains the responsibility of the users of the Standard to judge its suitability for their particular purpose.*

3) *This Standard was developed by consensus, which is defined by CSA Policy governing standardization — Code of good practice for standardization as "substantial agreement. Consensus implies much more than a simple majority, but not necessarily unanimity". It is consistent with this definition that a member may be included in the Technical Committee list and yet not be in full agreement with all clauses of this Standard.*

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d) *rationale for the change.*

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CAN/CSA-O15-15
Wood utility poles and reinforcing stubs

0 Introduction

This Standard covers all commercial pole species used in Canada. Since there is an exchange of poles between Canada, Europe, and the United States, this Standard also includes the imported species of Scots pine and southern yellow pine and harmonizes some of the physical attributes with other countries' standards.

During the development of the 2015 edition of this Standard, the Subcommittee agreed that published pole test data sets that are not included in Table A.1 of this Standard will be analyzed and, if deemed appropriate, the fibre stress and stiffness of the species in Table 3 will be updated accordingly in the next edition.

1 Scope

1.1

This Standard covers material, manufacturing, and class dimensional requirements for seasoned wood poles and reinforcing stubs that are primarily intended for the support of electrical power or communication lines.

1.2

This Standard does not cover requirements for wood poles intended for use in structures such as bridges and buildings.

1.3

This Standard does not cover requirements for the preservative treatment of poles or the conditioning of poles prior to treatment. Such requirements are detailed in the CAN/CSA-O80 Series of Standards unless otherwise specified.

1.4

The values given in SI units are the units of record for the purposes of this Standard. The values given in parentheses are for information and comparison only. Tables 1 to 23 are written in yard/pound units and equivalent SI (metric) versions are provided in Annex E.

1.5

In this Standard, “shall” is used to express a requirement, i.e., a provision that the user is obliged to satisfy in order to comply with the standard; “should” is used to express a recommendation or that which is advised but not required; and “may” is used to express an option or that which is permissible within the limits of the standard.

Notes accompanying clauses do not include requirements or alternative requirements; the purpose of a note accompanying a clause is to separate from the text explanatory or informative material.

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Notes to tables and figures are considered part of the table or figure and may be written as requirements.

Annexes are designated normative (mandatory) or informative (non-mandatory) to define their application.

2 Reference publications

This Standard refers to the following publications, and where such reference is made, it shall be to the edition listed below:

CSA Group

CAN/CSA-O80 Series-08 (R2012)
Wood preservation

ANSI (American National Standards Institute)

O5.TR.01-2009
Photographic Manual of Wood Pole Characteristics (Based on ANSI O5.1.2008)

ASTM International

D1036-99(2012)
Standard Test Methods of Static Tests of Wood Poles

Other publications

Natural Resources Canada. The Atlas of Canada.

U.S. Department of the Interior. National Atlas of the United States.
http://www.nationalatlas.gov/

3 Definitions

The following definitions apply in this Standard:

Bird holes — holes and damage caused by woodpeckers and other birds.

Check — a lengthwise separation of the wood that usually extends across the annual rings, commonly resulting from stresses set up in wood during seasoning.

Compression wood — abnormal wood usually formed on the lower side of branches and the concave side of leaning trees.

Note: *It is characterized by an unusually high proportion of apparent latewood in the annual rings, with a gradual transition from earlywood (springwood) to latewood (summerwood). It has a lifeless appearance and exhibits a darker colour when wetted. Compared to normal wood, compression wood shrinks excessively lengthwise.*

Crack —

> **Cross break crack** — a separation of the wood cells across the grain. Such breaks can occur when a tree is felled, during handling, or be due to internal strains resulting from unequal longitudinal shrinkage.

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Dead streak — a portion of the sapwood in which the life processes ceased prior to the cutting of the tree.

Note: *A dead streak differs from a wound such as a scar (cat face) in which the growth of new wood shows that life processes are still acting to repair the injured part. It is characterized by a discoloured, weathered appearance and a lack of growth of healthy wood overlapping the edges of the streak in an attempt to repair the damaged area.*

Decay (rot, dote, doze) — the deterioration of wood caused by the action of wood-destroying fungi, resulting in softening, loss of strength and weight, and often change of texture and colour.

Advanced decay — the late stage of decay in which the decomposition is readily recognized as the wood becomes punky, soft, stringy, pitted, or crumbly.

Incipient decay — the early stage of decay in which the decomposition has not proceeded far enough to soften or otherwise change the hardness of the wood perceptibly. It is usually accompanied by a slight discoloration or bleaching of the wood.

Note: *Laboratory analysis may be required for positive identification of incipient decay.*

Face of pole — the concave side of a pole having sweep in one plane or the side of greatest curvature between the groundline and the top of a pole having a reverse or double sweep.

Groundline section — that section of a pole between 0.305 m (1 ft) above and 0.61 m (2 ft) below the groundline as specified in the pole dimension tables of this Standard.

Hollow heart — a void in the heartwood caused by decay or insect attack.

Insect damage — damage that is the result of boring in the wood by insects or insect larvae (limits as defined in Clause 4.4.6).

Knot — that portion of a branch or limb that has been surrounded by a subsequent growth of wood of the trunk or other portion of the tree.

Dead knot — a knot resulting from a branch that died before the tree was cut.

Decayed knot (unsound knot) — a knot that is softer than the surrounding wood and shows evidence of advanced decay.

Knot cluster — two or more knots grouped together as a unit, with the fibres of the wood being deflected around the entire unit; a knot cluster is considered and measured as a single knot.

Ring knots — a group of knots that encircle a pole in the same plane; any knot intersecting an imaginary line drawn around a pole is included in the measurement of the sum of diameters of ring knots (see Table 1).

Spike knot — an elongated knot that results from a branch that grew on an angle not perpendicular to the trunk of a tree.

Note: *Spike knots are often encased in outer bark and/or tree sap, frequently resulting in a bark inclusion (limits as defined in Clause 4.4.8.2).*

Knot diameter — the dimension of a knot measured on the surface of a pole at a right angle to the lengthwise axis of the pole.

Note: *The sapwood as well as the heartwood portion of a knot are included in the measurement.*

Mechanical damage — surface damage incurred during the manufacturing or handling of poles.

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Plugged holes — holes that have been filled by forcing in a wood insert.

Red heart — a form of incipient decay, characterized by a reddish colour produced in the heartwood, that does not render the wood unfit for the majority of uses.

Note: *Firm red heart contains none of the white pockets that characterize the more advanced stage of decay. It is not to be confused with naturally red heartwood.*

Sap stain — a discoloration of the sapwood that is caused by the action of certain moulds and fungi and is not accompanied by softening or other deterioration of the wood.

Scar (cat face) — a depression in the surface of a pole resulting from a wound where healing has not re-established the normal cross-section of the pole.

Shake — separation along the grain, the greater part of which occurs between the annual growth rings.

Short crook — a localized deviation from straightness that, within any section 1.52 m (5 ft) or less in length, is more than one-half the mean diameter of the crooked section (see Figure 2).

Spiral grain (twisted grain) — a type of growth in which the fibres take a spiral course around the trunk of a tree instead of a vertical course.

Note: *The spiral can extend around the trunk in a right-hand or left-hand direction or can alternate right and left within the trunk. The degree of spirality in a pole is measured by the distance in feet along the axis of the pole in which one complete twist of the spiral occurs.*

Split — a separation along the grain forming an opening or fissure that extends through the piece from one surface to another.

Sweep — the deviation of a pole from straightness (see Figure 1).

4 Material requirements

4.1 General

4.1.1

The species covered by this Standard are listed in Table 3.

4.1.2

Poles and reinforcing stubs shall be classified in accordance with the circumferential dimensions specified in Tables 4 to 13 and 14 to 23, respectively. These dimensions were calculated from the classification stresses specified in Table 3 in accordance with the classification procedures described in Annexes B and C.

4.1.3

Species, length and class of poles, type of preservative treatment, and complete details for boring, gaining, and roofing shall be given to the supplier. Where requirements of this Standard are modified to meet special conditions, complete detailed instructions shall be given to the supplier. See Annex D for ordering information.

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4.2 Prohibited defects

All poles shall be free of the following defects:

a) bird holes;
b) cracks;
c) dead streaks, except as permitted in Clause 4.4.3;
d) decay, except as permitted by Clauses 4.4.4 and 4.4.5;
e) double sweep or reverse sweep in reinforcing stubs (see Clause 4.4.12);
f) plugged holes, except increment borer holes;
g) hollow heart in butts or tops, except as permitted by Clause 4.4.5;
h) marine borer damage;
i) nails, spikes, and other metal not specifically authorized by the purchaser;
j) short crook (see Clause 4.4.12); and
k) splits in the top.

4.3 Permitted defects

The following defects shall be permitted:

a) checks that result from normal seasoning conditions and that extend not deeper than the depth of pith (see Clause 4.4.1 for limitations on two or more checks that become contiguous at the pith);
b) fire-kill, provided that there is no char;
c) firm red heart; and
d) sap stain that is not accompanied by decay.

4.4 Limited defects

4.4.1 Contiguous checks, through-checks, and splits

Contiguous checks, through-checks, or splits in the butt surface shall be permitted, provided that their height from the butt along the side surface does not exceed 0.61 m (2 ft).

4.4.2 Compression wood

The outer 2.54 cm (1 in) of all poles shall be free of compression wood visible on either end.

4.4.3 Dead streaks

A single sound dead streak, not wider than one-quarter of the circumference of the pole measured at the widest point of the dead streak, shall be permitted in cedar poles only.

4.4.4 Decay in knots

Decay shall be permitted in knots, provided that it is not associated with heart rot.

4.4.5 Defective butts

Hollowing of the butt caused by “splinter pulling” in felling the tree shall be permitted, provided that the area of such a hollow is less than 10% of the butt area. Hollow heart and decay shall be permitted in cedar poles only, provided that the aggregate area of hollow heart and decay does not exceed 10% of the butt area and that the depth of the hollow does not exceed 0.61 m (2 ft), measured from the butt surface. Decay in the butt within 5.1 cm (2 in) of the pole surface shall be permitted in cedar poles only, provided that the total area of decay does not exceed 6.45 cm^2 (1 in^2).

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4.4.6 Insect damage

Insect damage consisting of holes 1.6 mm (1/16 in) or less in diameter shall be permitted. Surface scoring or channelling shall be permitted. All other forms of insect damage shall be prohibited.

4.4.7 Knot diameter

The diameter of any single knot or the sum of the diameters of all knots in any 0.305 m (1 ft) section shall not exceed the limits specified in Table 1. Knots 13 mm (1/2 in) or less in diameter shall be ignored in applying the limitations for the sum of diameters. The diameter of any single knot is measured at the widest point of the knot, perpendicular to the length of the pole, and includes the sapwood and heartwood of the knot, as shown in Figure 3.

4.4.8 Scars (cat faces)

4.4.8.1 Sound scars

Sound scars shall be permitted on a pole surface, provided that

a) they are not located within 0.61 m (2 ft) of the groundline;
b) they are smoothly trimmed and do not interfere with the cutting of any gain or drilled hole;
c) the circumference of any point on the trimmed surfaces located between the butt and 0.61 m (2 ft) below the groundline is not less than the minimum circumference specified at 1.83 m (6 ft) from the butt for the class and length of the pole;
d) the maximum width of the scar, after trimming, is not more than one-fifth of the pole circumference at that point; and
e) the maximum depth of the trimmed scar (bark removed) is not more than
    i) 25 mm (1 in) if the pole diameter at the location of the scar is 0.25 m (10 in) or less; or
    ii) one-tenth of the pole diameter at the location of the scar if such diameter is more than 0.25 m (10 in).

4.4.8.2 Bark inclusions

Sound bark inclusions, bark removed, shall be permitted on a pole surface, provided that

a) they are not located within 0.61 m (2 ft) of the groundline; and
b) the depth of the bark inclusion (bark removed) does not exceed
    i) one-fifth of the pole diameter at the bark inclusion if the inclusion is 20.3 cm (8 in) or less in length; or
    ii) 2.5 cm (1 in) if the inclusion is more than 20.3 cm (8 in) long.

4.4.8.3 Mechanical damage

Mechanical damage shall be permitted on a pole surface, provided that

a) they are not located within two feet of the groundline;
b) damage over 1.3 cm (½ in) in depth is smoothly trimmed and shaped to remove all broken and splintered fibre;
c) the circumference of any point on the trimmed surfaces located between the butt and 0.61 m (2 ft) below the groundline is not less than the minimum circumference specified at 6 ft from the butt for the class and length of the pole;
d) the maximum width of the damage, after trimming, is not more than one-fifth of the pole circumference at that point; and
e) the maximum depth of the trimmed damage is not more than
    i) 25 mm (1 in) if the pole diameter at the location of the scar is 0.25 m (10 in) or less; or

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ii) one-tenth of the pole diameter at the location of the damage if such diameter is more than 0.25 m (10 in).

4.4.9 Shakes in butts

Shakes in the butt end surface extending through an arc of not more than 90° shall be permitted. Shakes extending through an arc of more than 90° shall be permitted if they are inside a circle, the centre of which corresponds to the centre of the butt surface and the diameter of which corresponds to one-half of the average butt diameter.

4.4.10 Shakes in tops

Shakes in the top end surface shall be permitted in poles that are to be given full-length preservative treatment, provided that the diameter of the shake does not exceed one-half of the diameter of the top of the pole.

4.4.11 Spiral grain

Spiral grain (twist grain) shall be permitted within the limits specified in Table 2.

①

4.4.12 Crooks and sweep

Crooks other than short crooks shall be permitted in poles. Sweep shall be permitted, provided that a straight line connecting the midpoint at the groundline with the midpoint at the top shall not, at any intermediate point, pass through the surface of the pole. Double or reverse sweep shall not be permitted in reinforcing stubs. (See Figures 1 and 2.)

4.4.13 Rate of growth

4.4.13.1

Except as specified in Clause 4.4.13.2, the rate of growth measured on the butt in the outer 51 mm (2 in) of poles having a circumference of 0.95 m (37.5 in) or less at 1.83 m (6 ft) from the butt, and in the outer 76 mm (3 in) of poles having a circumference of more than 0.95 m (37.5 in) at 1.83 m (6 ft) from the butt, shall be not less than 6 rings per 2.54 cm (1 in).

4.4.13.2

Poles with 4 and 5 rings per 2.54 cm (1 in) shall be acceptable if 50% or more summerwood is present.

4.4.13.3

As an alternative to Clause 4.4.13.1, the ring count and summerwood measurements may be made on an increment core taken at 1.83 m (6 ft) from the butt directly above the place where the average growth is indicated on the butt surface.

Note: *ANSI O5.TR.01 provides accurate photographic references for the limited and prohibited defects noted in this Standard.*

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5 Dimensions

5.1 Poles

5.1.1 General

Poles shall be classified according to species, length, and minimum circumferential measurements as specified in Tables 4 to 13.

5.1.2 Length

Poles 45 ft in length or shorter shall be not more than 76 mm (3 in) shorter or 76 mm (3 in) longer than the nominal length. Poles 15.24 m (50 ft) or more in length shall be not more than 76 mm (3 in) shorter or 152 mm (6 in) longer than the nominal length. Length shall be measured between the extreme ends of the pole.

5.1.3 Circumference

The minimum circumferences at 1.83 m (6 ft) from the butt and at the top, for each length and class of pole, are listed in Tables 4 to 13. The circumference at 1.83 (6 ft) from the butt of a pole shall be not more than 178 mm (7 in) or 20% larger than the specified minimum, whichever is greater. The top circumferential requirement shall apply at a point corresponding to the minimum length permitted for the pole.

5.1.4 Classification method

After the species and length of a pole are determined, its class shall be determined by measuring circumferences at two cross-sections: the top and 1.83 m (6 ft) from the butt. The latter dimension shall determine the class of the pole by reference to the applicable table in Tables 4 to 13, provided that its top (measured at the minimum length point) is large enough. Otherwise, the circumference at the top shall determine the class, provided that the circumference at 1.83 m (6 ft) from the butt does not exceed the specified minimum by more than 178 mm (7 in) or 20%, whichever is greater.

5.2 Reinforcing stub circumference

The minimum circumference at 1.83 m (6 ft) from the butt, for each length and class of stub, shall be as specified in Tables 14 to 23.

6 Quality of work and finish, markings, handling, storage, and inspection

6.1 Bark removal

6.1.1

The outer bark shall be completely removed from all poles. No patch of inner bark more than 25 mm (1 in) wide and 152 mm (6 in) long shall be left on the pole surface between the top and 0.61 m (2 ft) below the groundline.

6.1.2

In shaving poles, the depth of cut shall be not more than necessary to remove inner bark. There shall be no abrupt changes in the contour of the pole surface between the groundline and top sections.

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6.2 Sawing

Poles shall be sawed neatly at the top and at the butt along planes that shall not be out of square with the axis of the pole by more than 10% of the diameter of the sawed surface. There may be bevelling at the edge of the sawed butt surface not wider than 10% of the butt diameter.

6.3 Trimming

Completely overgrown knots rising more than 25 mm (1 in) above the pole surface, branch stubs, and partially overgrown knots shall be trimmed close. Completely overgrown knots less than 25 mm (1 in) high need not be trimmed.

6.4 Framing

All poles shall be roofed, gained, and bored as specified by the purchaser before full-length treatment. All gains of a notch or slab type shall be cut on the face of the pole, and the gained surfaces shall be in parallel planes.

6.5 Markings

6.5.1 Butt marking

The following marks shall be stamped or branded legibly on the butt end of each pole or be included on a metal tag affixed thereto:

a) the supplier's code or trademark;
b) code letters denoting the pole species (see Clause 6.5.3); and
c) the class and length of the pole.

Marks stamped or branded on the butt end shall be not less than 16 mm (5/8 in) high. Marks on metal tags shall be not less than 3.2 mm (1/8) in high.

6.5.2 Side marking

6.5.2.1

Not more than one month before treatment, the following marks, not less than 16 mm (5/8) in high, shall be branded legibly and permanently on the face of each pole:

a) the supplier's code or trademark;
b) the plant location and the last two numerals of the year of treatment;
c) code letters denoting the country of origin (for imported poles) and pole species (see Clause 6.5.3);
d) the class and length of the pole; and
e) following the species code letters [see Item c)], the appropriate code letter to denote the type of preservative, as specified in the CAN/CSA-O80 Series of Standards.

Note: *Subject to agreement between the supplier and purchaser, die-stamped metal tags affixed to the pole face may be used in place of brands to designate side marking information. If metal tags are used, the code letters specified in Item c) should be not less than 1/8 in high.*

6.5.2.2

The bottom of the side mark shall be located on the face of the pole at 3.05 m ± 0.05 m (10 ft ± 2 in) from the butt of poles 15.2 m (50 ft) or less in length, and at 4.27 m ± 0.05 m (14 ft ± 2 in) from the butt of poles 16.8 m (55 ft) or more in length. In the case of 3.05 m (10 ft), 3.35 m (11 ft), and 3.96 m (13 ft) stubs, the mark should be 0.31 m (1 ft) from the top.

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6.5.3 Code letters for country of origin and pole species

The following code letters shall be used to designate the country of origin of an imported pole and the species of pole. The code letter for the country of origin shall precede the code letters for the species on all imported poles except those of southern yellow pine:

Country of origin	Code letter	Species	Code letters
Finland	F-	Cedar	
Norway	N-	Eastern white	EC
Sweden	S-	Western red	WC
United States	U-	Yellow	YC
		Fir	
		Amabilis	AF
		Douglas	DF
		Hemlock	
		Western	WH
		Larch	
		Western	WL
		Pine	
		Eastern white	EP
		Jack	JP
		Lodgepole	LP
		Ponderosa	PP
		Red	RP
		Scots	SCP
		Southern yellow	SP
		Spruce	
		Black	WS
		Engelmann	ES
		Red	WS
		Sitka	SS
		White	WS

6.6 Handling of poles and reinforcing stubs

6.6.1

Treated material shall not be dragged along the ground.

6.6.2

Cant hooks, pole tongs, or other tools shall not be applied to the groundline section of any pole.

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6.6.3
Loading irons and cant hooks shall not be used on the roofed surface of any pole.

6.6.4
Poles shall not be dropped or otherwise roughly handled.

6.7 Storage
When it is necessary to store poles, they shall be stacked on treated skids of such dimensions and arranged so as to support the poles without producing noticeable distortion in any of them. The height of the piles shall be limited to avoid damage to poles in the bottom layers. Poles shall be piled and supported in such a manner that all poles are at least 0.31 m (1 ft) above the general ground level. No vegetation or decaying wood shall be permitted underneath stored poles.

6.8 Inspection
All poles may be subject to inspection by inspectors appointed by the purchaser. Such inspectors shall have the power to reject any pole that does not comply with the requirements of this Standard.

Table 1
Maximum knot sizes
(See Clauses 3, 4.4.7 and E.1.)

	Maximum knot sizes		
	Diameter of any single knot, in		Sum of diameters of all knots in any length of 1 ft
Length of pole, ft	Classes H6 to 3	Classes 4 to 7	All classes
45 and shorter	4	3	One-third of the average circumference of the same 1 ft section or 8 in, whichever is greater, but not to exceed 12 in
50 and longer	5	5	One-third of the average circumference of the same 1 ft section or 10 in, whichever is greater, but not to exceed 14 in

Table 2
Maximum spiral grain
(See Clauses 4.4.11 and E.1.)

Length of pole, ft	Maximum twist of grain permitted
30 and shorter	1 complete twist in any 10 ft length
31– 49	1 complete twist in any 16 ft length
50 and longer	1 complete twist in any 20 ft length

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Table 3
Strength and stiffness properties by species
(See Clauses 4.1.1, 4.1.2, B.1.1, and E.1 and Tables 5, 7, 10, 11, 13, and 23.)

Common name	Botanical name	Fibre stress, psi	Fibre stress coefficient of variation, %	Modulus of elasticity, 1000 psi	Modulus of elasticity coefficient of variation, %
Cedar					
Eastern white	*Thuja occidentalis*	3700	18	—	—
Western red	*Thuja plicata*	5600	19	1120	14
Yellow	*Chamaecyparis nootkatensis*	6600	—	—	—
Fir					
Douglas	*Pseudotsuga menziesii*				
Coastal	—	8000	15	2030	11
Interior	—	6800	15	—	—
Amabilis	*Abies amabilis*	5400	18	1220	19
Hemlock					
Western	*Tsuga heterophylla*	6800	17	1390	15
Larch					
Western	*Larix occidentalis*	8000	17	1885	20
Pine					
Eastern white	*Pinus strobus*	4900	—	—	—
Jack	*Pinus banksiana*	6400	20	—	—
Lodgepole	*Pinus contorta*	6600	13	1190	14
Ponderosa	*Pinus ponderosa*	5600	—	—	—

(Continued)

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Table 3 (Continued)

Common name	Botanical name	Fibre stress, psi	Fibre stress coefficient of variation, %	Modulus of elasticity, 1000 psi	Modulus of elasticity coefficient of variation, %
Red	*Pinus resinosa*	6000	17	1290	14
Scots	*Pinus sylvestris*	7800	14	1520	18
Southern yellow	—	7800	24	1595	29
Spruce	—	4900	17	1030	17

Notes:

1) *The fibre stresses are stresses for the purposes of classification and are close approximations to the average bending strengths (MOR) specified in Table A.1 for poles 20 to 35 ft in length.*

2) *The bending strengths specified in Table A.1 are derived from specimens whose moisture contents at the groundline were above the fibre saturation point at the time of testing to failure. This condition is recognized as the best approximation for the purposes of this Standard. Furthermore, the average moduli of elasticity specified in this Table are derived from poles of any length that were unseasoned or completely submerged in water for some time before testing, as described in the publications referenced in Clause A.9.*

3) *The strength of poles longer than 50 ft can be less, as indicated in Annex A.*

4) *Fibre stress for several species was based on limited test information or results from one test site; therefore, the Subcommittee is uncertain of the adequacy of the pole dimensions for these species and recommends that further tests be performed to supplement the database. Notes concerning fibre stress accompany Tables 3 and E.3.*

5) *Other criteria related to the classification of poles are as follows:*

a) *The taper for determining the circumference at the classification point should be the difference between the minimum top circumference and the circumference at the theoretical groundline, divided by the length between them.*

b) *The loading point should be 0.61 m (2 ft) from the pole top.*

c) *The classification point should be 1.83 m (6 ft) from the butt.*

d) *The term "class" as used in this Standard does not refer to quality limitations or defects of a pole or reinforcing stub; it is used to differentiate poles and stubs on the basis of their potential force-resisting capabilities.*

6) *Pole test data for yellow cedar, eastern white pine, and ponderosa pine were not available; hence, the values specified in this Table approximate the average strengths of small, clear bending specimens, as described in the publications referenced in Clause A.9.*

7) *Samples of Douglas fir poles taken from coastal areas of British Columbia and from the Cascade Mountains region of Oregon have significantly higher average strengths than those taken from Rocky Mountain or interior areas. Thus, only Douglas fir poles taken from the western slopes of the Coast and Cascade Mountains (as defined by the Atlas of Canada and the National Atlas of the United States) shall be classified as coastal Douglas fir. All other Douglas fir poles shall be classified as interior Douglas fir.*

(Continued)

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Table 3 (Concluded)

8) *In addition to the natural variability within a sample of poles, there is evidence from the data in Annex A that there is significant variability between pole samples taken from various areas of its growth region.*

9) *Scots and southern yellow pines are not native to Canada. Southern yellow pine is the name for a commercial grouping of species that includes loblolly (Pinus taeda), longleaf (P. palustris), shortleaf (P. echinata), and slash (P. elliottii) pines.*

10) *Spruce species include black (Picea mariana), Engelmann (P. engelmannii), red (P. rubens), Sitka (P. sitchensis), and white (P. glauca) spruce, of which only the latter has been tested in a pole size. Separation of red and black spruce from white spruce is impractical once the pole is removed from the forest. Small, clear bending specimens tested for each species indicate that white spruce is the weakest of the five.*

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Table 4
Dimensions of eastern white cedar poles (fibre stress 3700 psi)
(See Clauses 4.1.2, 5.1.1, 5.1.3, 5.1.4, B.1.1, B.2, and E.1.)

Class		1	2	3	4	5	6	7
Minimum circumference at top, in		27	25	23	21	19	17	15
Length of pole, ft	Groundline distance from butt, ft*	Minimum circumference at 6 ft from butt, in						
20	4	38.5	36.0	33.5	31.0	28.5	26.5	24.5
25	5	43.0	40.0	37.5	34.5	32.0	29.5	27.5
30	5.5	46.5	43.5	40.5	38.0	35.0	32.5	30.0
35	6	50.0	47.0	43.5	40.5	37.5	34.5	32.0
40	6	53.0	49.5	46.0	43.0	39.5	36.5	34.0
45	6.5	55.5	52.0	48.5	45.0	41.5	38.5	—
50	7	58.0	54.5	51.0	47.0	43.5	—	—
55	7.5	60.5	56.5	53.0	49.0	—	—	—
60	8	62.5	58.5	54.5	51.0	—	—	—
65	8.5	64.5	60.5	56.5	52.5	—	—	—

* *These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

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Table 5
Dimensions of spruce and eastern white pine poles (fibre stress 4900 psi)

(See Clauses 4.1.2, 5.1.1, 5.1.3, 5.1.4,B.1.1, B.2, and E.1.)

Class		1	2	3	4	5	6	7
Minimum circumference at top, in		27	25	23	21	19	17	15
Length of pole, ft	Groundline distance from butt, ft*	Minimum circumference at 6 ft from butt, in						
20	4	35.5	33.0	31.0	28.5	26.5	24.5	22.5
25	5	39.0	36.5	34.0	31.5	29.5	27.0	25.0
30	5.5	42.5	40.0	37.0	34.5	32.0	29.5	27.5
35	6	45.5	42.5	39.5	37.0	34.0	31.5	29.5
40	6	48.0	45.0	42.0	39.0	36.0	33.5	31.0
45	6.5	50.5	47.5	44.0	41.0	38.0	35.0	32.5
50	7	53.0	49.5	46.0	43.0	39.5	36.5	—
55	7.5	55.0	51.5	48.0	44.5	41.5	38.0	—
60	8	57.0	53.5	50.0	46.0	—	—	—
65	8.5	59.0	55.0	51.5	47.5	—	—	—

* *These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

Note: *See the Note 2) to Table 3 on eastern white pine.*

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Table 6
Dimensions of amabilis fir poles (fibre stress 5400 psi)
(See Clauses 4.1.2, 5.1.1, 5.1.3, 5.1.4, B.1.1, B.2, and E.1.)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7
Minimum circumference at top, in		**39**	**37**	**35**	**33**	**31**	**29**	**27**	**25**	**23**	**21**	**19**	**17**	**15**
Length of pole, ft	**Groundline distance from butt, ft***	**Minimum circumference at 6 ft from butt, in**												
20	4	—	—	—	—	—	—	34.5	32.0	30.0	27.5	25.5	23.5	22.0
25	5	—	—	—	—	—	—	38.0	35.5	33.0	30.5	28.5	26.0	24.5
30	5.5	—	—	—	—	—	—	41.0	38.5	36.0	33.5	31.0	28.5	26.5
35	6	—	—	—	—	49.5	47.0	44.0	41.0	38.5	35.5	33.0	30.5	28.5
40	6	—	—	58.0	55.0	52.5	49.5	46.5	43.5	40.5	38.0	35.0	32.5	—
45	6.5	66.5	64.0	61.0	58.0	55.0	52.0	49.0	46.0	43.0	39.5	36.5	34.0	—
50	7	69.5	66.5	63.5	60.5	57.5	54.5	51.0	48.0	44.5	41.5	38.5	—	—
55	7.5	72.5	69.5	66.0	63.0	60.0	56.5	53.0	50.0	46.5	43.0	—	—	—
60	8	75.0	72.0	68.5	65.5	62.0	58.5	55.0	51.5	48.0	44.5	—	—	—
65	8.5	77.5	74.0	71.0	67.5	64.0	60.5	57.0	53.5	49.5	46.0	—	—	—
70	9	80.0	76.5	73.0	69.5	66.0	62.5	58.5	55.0	51.0	47.5	—	—	—
75	9.5	82.0	78.5	75.0	71.5	67.5	64.0	60.0	56.5	52.5	—	—	—	—
80	10	84.0	80.5	77.0	73.0	69.5	65.5	62.0	58.0	54.0	—	—	—	—
85	10.5	86.0	82.5	78.5	75.0	71.0	67.0	63.0	59.5	55.5	—	—	—	—
90	11	88.0	84.5	80.5	76.5	72.5	68.5	64.5	60.5	56.5	—	—	—	—
95	11	89.5	86.0	82.0	78.0	74.0	70.0	66.0	62.0	—	—	—	—	—

(Continued)

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Table 6 (Concluded)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7
Minimum circumference at top, in		**39**	**37**	**35**	**33**	**31**	**29**	**27**	**25**	**23**	**21**	**19**	**17**	**15**
Length of pole, ft	**Groundline distance from butt, ft***	**Minimum circumference at 6 ft from butt, in**												
100	12	91.5	87.5	83.5	79.5	75.5	71.5	67.0	63.0	—	—	—	—	—
105	12	93.5	89.5	85.5	81.0	77.0	73.0	68.5	64.0	—	—	—	—	—
110	12	95.0	91.0	86.5	82.5	78.5	74.0	69.5	65.5	—	—	—	—	—
115	12	96.5	92.0	88.0	84.0	79.5	75.0	70.5	66.5	—	—	—	—	—
120	12	97.5	93.5	89.5	85.0	80.5	76.5	72.0	67.5	—	—	—	—	—
125	12	99.0	95.0	90.5	86.5	82.0	77.5	73.0	68.0	—	—	—	—	—

** These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, et*

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Table 7
Dimensions of western red cedar and ponderosa pine poles (fibre stress 5600 psi)

(See Clauses 4.1.2, 5.1.1, 5.1.3, 5.1.4, B.1.1, B.2, and E.1.)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7
Minimum circumference at top, in		39	37	35	33	31	29	27	25	23	21	19	17	15
Length of pole, ft	Groundline distance from butt, ft*	Minimum circumference at 6 ft from butt, in												
20	4	—	—	—	—	—	—	33.5	31.5	29.5	27.0	25.0	23.0	21.5
25	5	—	—	—	—	—	—	37.0	34.5	32.5	30.0	28.0	25.5	24.0
30	5.5	—	—	—	—	—	—	40.0	37.5	35.0	32.5	30.0	28.0	26.0
35	6	—	—	—	—	48.0	45.5	42.5	40.0	37.5	34.5	32.0	30.0	27.5
40	6	—	—	56.5	53.5	51.0	48.0	45.0	42.5	39.5	36.5	34.0	31.5	—
45	6.5	64.5	62.0	59.0	56.0	53.5	50.5	47.5	44.5	41.5	38.5	36.0	33.0	—
50	7	67.0	64.5	61.5	58.5	55.5	52.5	49.5	46.5	43.5	40.0	37.5	—	—
55	7.5	70.0	67.0	64.0	61.0	57.5	54.5	51.5	48.5	45.0	42.0	—	—	—
60	8	72.0	69.0	66.0	63.0	59.5	56.5	53.5	50.0	46.5	43.5	—	—	—
65	8.5	74.5	71.5	68.0	65.0	61.5	58.5	55.0	51.5	48.0	45.0	—	—	—
70	9	76.5	73.5	70.0	67.0	63.5	60.0	56.5	53.0	49.5	46.0	—	—	—
75	9.5	78.5	75.5	72.0	68.5	65.0	61.5	58.0	54.5	51.0	—	—	—	—
80	10	80.5	77.0	74.0	70.5	67.0	63.0	59.5	56.0	52.0	—	—	—	—
85	10.5	82.5	79.0	75.5	72.0	68.5	64.5	61.0	57.0	53.5	—	—	—	—
90	11	84.5	81.0	77.0	73.5	70.0	66.0	62.5	58.5	54.5	—	—	—	—
95	11	86.0	82.5	79.0	75.0	71.5	67.5	63.5	59.5	—	—	—	—	—

(Continued)

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Table 7 (Concluded)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7
Minimum circumference at top, in		39	37	35	33	31	29	27	25	23	21	19	17	15
Length of pole, ft	Groundline distance from butt, ft*	Minimum circumference at 6 ft from butt, in												
100	11	87.5	84.0	80.5	76.5	72.5	68.0	65.0	61.0	—	—	—	—	—
105	12	89.5	85.5	82.0	78.0	74.0	70.0	66.0	62.0	—	—	—	—	—
110	12	91.0	87.0	83.5	79.5	75.5	71.5	67.5	63.0	—	—	—	—	—
115	12	92.5	88.5	84.5	80.5	76.5	72.5	68.5	64.0	—	—	—	—	—
120	12	94.0	90.0	86.0	82.0	78.0	74.0	69.5	65.0	—	—	—	—	—
125	12	95.0	91.5	87.5	83.0	79.0	75.0	70.5	66.0	—	—	—	—	—

** These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

Notes:

1) *For marketing purposes, yellow cedar can be considered the same as western red cedar on a dimensional basis only, but such poles shall be identified by actual species.*

2) *See Note 2) to Table 3 on Ponderosa pine.*

3) *The loads specified in Table B.1 should be corrected by a factor of 0.95 for western red cedar.*

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Table 8
Dimensions of red pine poles (fibre stress 6000 psi)
(See Clauses 4.1.2, 5.1.1, 5.1.3, 5.1.4, B.1.1, B.2, and E.1.)

Class		1	2	3	4	5	6	7	8
Minimum circumference at top, in		27	25	23	21	19	17	15	15
Length of pole, ft	Groundline distance from butt, ft*	Minimum circumference at 6 ft from butt, in							
20	4	32.5	30.5	28.5	26.5	24.5	22.5	21.0	20.0
25	5	36.0	33.5	31.0	29.0	27.0	25.0	23.0	22.0
30	5.5	39.0	36.5	34.0	31.5	29.0	27.0	25.0	24.0
35	6	41.5	38.5	36.0	33.5	31.0	28.5	26.5	25.5
40	6	44.0	41.0	38.0	35.5	33.0	30.5	—	—
45	6.5	46.0	43.0	40.0	37.0	34.5	32.0	—	—
50	7	48.0	45.0	42.0	39.0	36.0	—	—	—
55	7.5	49.5	46.5	43.5	40.5	—	—	—	—
60	8	51.5	48.0	45.0	42.0	—	—	—	—
65	8.5	53.0	49.5	46.0	43.0	—	—	—	—

** These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

Notes:

1) *There is evidence to suggest that red pine poles of average strength lower than 6000 psi can be obtained from certain areas of the United States.*

2) *The loads specified in Table B.1 should be corrected by a factor of 0.95 for red pine.*

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Gst}wklxGWEK~syt$
Hsgyq irx$t~vszmih$f}LWyrhirx$pmgirwi$jsqGWEK~syt
Rs$vitvshyqpmsr$sv$rix{svqmrk$tivqmxih$(mdsyx$pmgirwijsqLW
Pmgirwi AQ ermsfe$L}hvs9=4<5==5440$Vwiv A[s{vjo0$Newsr
RsxjsvViwepi0$4<35<3645:$56:98=$Q HX

Table 9
Dimensions of Jack pine poles (fibre stress 6400 psi)
(See Clauses 4.1.2, 5.1.1, 5.1.3, 5.1.4, B.1.1, B.2, and E.1.)

Class		1	2	3	4	5	6	7	8
Minimum circumference at top, in		**27**	**25**	**23**	**21**	**19**	**17**	**15**	**15**
Length of pole, ft	**Groundline distance from butt, ft***	**Minimum circumference at 6 ft from butt, in**							
20	4	32.5	30.5	28.5	26.5	24.5	22.5	21.0	20.0
25	5	36.0	33.5	31.0	29.0	27.0	25.0	23.0	22.0
30	5.5	39.0	36.5	34.0	31.5	29.0	27.0	25.0	24.0
35	6	41.5	38.5	36.0	33.5	31.0	28.5	26.5	25.5
40	6	44.0	41.0	38.0	35.5	33.0	30.5	—	—
45	6.5	46.0	43.0	40.0	37.0	34.5	32.0	—	—
50	7	48.0	45.0	42.0	39.0	36.0	—	—	—
55	7.5	49.5	46.5	43.5	40.5	—	—	—	—
60	8	51.5	48.0	45.0	42.0	—	—	—	—
65	8.5	53.0	49.5	46.0	43.0	—	—	—	—

** These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

Note: *The loads specified in Table B.1 should be corrected by a factor of 0.95 for Jack pine.*

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Gst}vklxGWEKvsyt$
Hsgyqirx$tvszmih$F}9LW$yrhivfpgirwi$jsvqGWEKvsyt
Rs$vitvshyqpsr$sv$rix{svomrk$tivqmxih$(mdsyx$pgirwi$jsvq$9LW
Pmgirwi AQ ermsfe$L}hvs9=4<5==5440$Vwiv A[s{vjo0$Newsr
RsxjsvViwepi0$4<35<3645:$56:98=$Q HX

Table 10
Dimensions of yellow cedar and lodgepole pine poles (fibre stress 6600 psi)

(See Clauses 4.1.2, 5.1.1, 5.1.3, 5.1.4, B.1.1, B.2, and E.1.)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7	8
Minimum circumference at top, in		39	37	35	33	31	29	27	25	23	21	19	17	15	15
Length of pole, ft	Groundline distance from butt, ft*	Minimum circumference at 6 ft from butt, in													
20	4	—	—	—	—	—	—	32.5	30.5	28.5	26.5	24.5	22.5	21.0	19.-5
25	5	—	—	—	—	—	—	36.0	33.5	31.0	29.0	27.0	25.0	23.0	21.-5
30	5.5	—	—	—	—	—	—	39.0	36.5	34.0	31.5	29.0	27.0	25.0	23.-0
35	6	—	—	—	—	46.5	43.5	41.5	38.5	36.0	33.5	31.0	28.5	26.5	25.-0
40	6	—	—	54.5	51.5	49.0	46.5	44.0	41.0	38.0	35.5	33.0	30.5	—	—
45	6.5	62.5	59.5	57.0	54.0	51.5	48.5	46.0	43.0	40.0	37.0	34.5	32.0	—	—
50	7	65.0	62.5	59.5	56.5	53.5	51.0	48.0	45.0	42.0	39.0	36.0	—	—	—
55	7.5	67.5	65.0	62.0	59.0	56.0	53.0	49.5	46.5	43.5	40.5	—	—	—	—
60	8	70.0	67.0	64.0	61.0	58.0	54.5	51.5	48.0	45.0	42.0	—	—	—	—
65	8.5	72.5	69.5	66.0	63.0	59.5	56.5	53.0	49.5	46.0	43.0	—	—	—	—
70	9	74.5	71.5	68.0	65.0	61.5	58.0	54.5	51.0	47.5	44.5	—	—	—	—
75	9.5	76.5	73.5	70.0	66.5	63.0	59.5	56.0	52.5	49.0	—	—	—	—	—
80	10	78.5	75.0	72.0	68.5	65.0	61.5	57.5	54.0	50.5	—	—	—	—	—

(Continued)

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Table 10 (Concluded)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7	8
Minimum circumference at top, in		39	37	35	33	31	29	27	25	23	21	19	17	15	15
Length of pole, ft	Groundline distance from butt, ft*	Minimum circumference at 6 ft from butt, in													
85	10.5	80.5	77.0	73.5	70.0	66.5	62.5	58.5	55.0	51.5	—	—	—	—	—
90	11	82.0	78.5	75.0	71.5	68.0	64.0	60.0	56.5	52.5	—	—	—	—	—
95	11	84.0	80.0	76.5	73.0	69.0	65.5	61.5	57.5	—	—	—	—	—	—
100	11	85.5	81.5	78.0	74.5	70.5	66.5	62.5	58.5	—	—	—	—	—	—
105	12	87.0	83.5	79.5	76.0	72.0	68.0	63.5	60.0	—	—	—	—	—	—
110	12	88.5	84.5	81.0	77.0	73.0	69.0	65.0	61.0	—	—	—	—	—	—
115	12	90.0	86.0	82.0	78.5	74.0	70.0	66.0	62.0	—	—	—	—	—	—
120	12	91.5	87.5	83.5	79.5	75.5	71.0	67.0	63.0	—	—	—	—	—	—
125	12	92.5	88.5	84.5	80.5	76.5	72.5	68.0	64.0	—	—	—	—	—	—

** These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

Notes:

1) *For marketing purposes, yellow cedar can be considered the same as western red cedar on a dimensional basis only, but such poles shall be identified by actual species.*

2) *See Note 2) to Table 3 on yellow cedar.*

3) *Lodgepole pine poles will not normally include Classes H1 to H6, nor be taller than 50 ft.*

4) *There is evidence to suggest that lodgepole pine poles of average strength lower than 6600 psi can be obtained from certain areas of the United States.*

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Table 11
Dimensions of western hemlock and interior Douglas fir poles (fibre stress 6800 psi)

(See Clauses 4.1.2, 5.1.1, 5.1.3, 5.1.4, B.1.1, B.2, and E.1.)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7
Minimum circumference at top, in		39	37	35	33	31	29	27	25	23	21	19	17	15
Length of pole, ft	Groundline distance from butt, ft*	Minimum circumference at 6 ft from butt, in												
20	4	—	—	—	—	—	—	32.0	30.0	28.0	26.0	24.0	22.0	20.5
25	5	—	—	—	—	—	—	35.0	33.0	30.5	28.5	26.5	24.5	22.5
30	5.5	—	—	—	—	—	—	38.0	35.5	33.5	31.0	28.5	26.5	24.5
35	6	—	—	—	—	46.0	43.5	41.0	38.0	35.5	33.0	30.5	28.5	26.0
40	6	—	—	53.5	51.0	48.5	46.0	43.0	40.5	37.5	35.0	32.5	30.0	—
45	6.5	61.5	59.0	56.5	53.5	51.0	48.0	45.5	42.5	39.5	37.0	34.0	31.5	—
50	7	64.5	61.5	59.0	56.0	53.0	50.5	47.5	44.5	41.5	38.5	35.5	—	—
55	7.5	67.0	64.0	61.0	58.5	55.5	52.5	49.0	46.0	43.0	40.0	—	—	—
60	8	69.5	66.5	63.5	60.5	57.5	54.0	51.0	47.5	44.5	41.5	—	—	—
65	8.5	71.5	68.5	65.5	62.5	59.0	56.0	52.5	49.5	46.0	42.5	—	—	—
70	9	73.5	70.5	67.5	64.0	61.0	57.5	54.0	51.0	47.5	44.0	—	—	—
75	9.5	76.0	72.5	69.5	66.0	62.5	59.0	55.5	52.0	48.5	—	—	—	—
80	10	77.5	74.5	71.0	67.5	64.0	60.5	57.0	53.5	50.0	—	—	—	—
85	10.5	79.5	76.0	72.5	69.0	65.5	62.0	58.5	55.0	51.0	—	—	—	—
90	11	81.5	78.0	74.5	71.0	67.0	63.5	60.0	56.0	52.0	—	—	—	—
95	11	83.0	79.5	76.0	72.0	68.5	64.5	61.0	57.0	—	—	—	—	—

(Continued)

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Table 11 (Concluded)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7
Minimum circumference at top, in		**39**	**37**	**35**	**33**	**31**	**29**	**27**	**25**	**23**	**21**	**19**	**17**	**15**
Length of pole, ft	**Groundline distance from butt, ft***	**Minimum circumference at 6 ft from butt, in**												
100	11	84.5	81.0	77.0	73.5	70.0	66.0	62.0	58.0	—	—	—	—	—
105	12	86.0	82.5	79.0	75.0	71.0	67.5	63.5	59.5	—	—	—	—	—
110	12	87.5	84.0	80.0	76.5	72.5	68.5	64.5	60.5	—	—	—	—	—
115	12	89.0	85.0	81.5	77.5	73.5	69.5	65.5	61.5	—	—	—	—	—
120	12	90.5	86.5	82.5	78.5	74.5	70.5	66.5	62.0	—	—	—	—	—
125	12	91.5	88.0	84.0	80.0	75.5	71.5	66.5	63.0	—	—	—	—	—

* *These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

Note: *See Note 3) to Table 3 for the classification of interior Douglas fir.*

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Table 12
Dimensions of southern yellow and Scots pine poles (fibre stress 7800 psi)
(See Clauses 4.1.2, 5.1.1, 5.1.3, 5.1.4, B.1.1, B.2, and E.1.)

Class		1	2	3	4	5	6	7
Minimum circumference at top, in		27	25	23	21	19	17	15
Length of pole, ft	Groundline distance from butt, ft*	Minimum circumference at 6 ft from butt, in						
20	4	31.0	29.0	27.0	25.0	23.0	21.0	19.5
25	5	33.5	31.5	29.5	27.5	25.5	23.0	21.5
30	5.5	36.5	34.0	32.0	29.5	27.5	25.0	23.5
35	6	39.0	36.5	34.0	31.5	29.0	27.0	25.0
40	6	41.0	38.5	36.0	33.5	31.0	28.5	—
45	6.5	43.5	40.5	37.5	35.0	32.5	30.0	—
50	7	45.0	42.0	39.0	36.5	34.0	—	—
55	7.5	46.5	43.5	40.5	38.0	—	—	—
60	8	48.0	45.0	42.0	39.0	—	—	—
65	8.5	49.5	46.5	43.5	40.5	—	—	—

** These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

Note: *For southern yellow pine, the average number of annual growth rings measured on the butt in the outer 2 in of poles having a circumference of 37.5 in or less at 6 ft from the butt, and in the outer 3 in of poles having a circumference of more than 3 ft at 6 ft from the butt, shall be not less than 12 and 18 rings, respectively. However, pole butts with 8 to 10 and 12 to 15 rings, respectively, shall be acceptable if 50% or more latewood is present. As an alternative, these ring count and latewood measurements may be made on an increment core taken at 6 ft from the butt directly above the place where the average number of annual rings is indicated on the butt surface.*

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Table 13
Dimensions of western larch and coastal Douglas fir poles (fibre stress 8000 psi)

(See Clauses 4.1.2, 5.1.1, 5.1.3, 5.1.4, B.1.1, B.2, and E.1.)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7
Minimum circumference at top, in		39	37	35	33	31	29	27	25	23	21	19	17	15
Length of pole, ft	Groundline distance from butt, ft*	Minimum circumference at 6 ft from butt, in												
20	4	—	—	—	—	—	—	31.0	29.0	27.0	25.0	23.0	21.0	19.5
25	5	—	—	—	—	—	—	33.5	31.5	29.5	27.5	25.5	23.0	21.5
30	5.5	—	—	—	—	—	—	36.5	34.0	32.0	29.5	27.5	25.0	23.5
35	6	—	—	—	—	43.5	41.5	39.0	36.5	34.0	31.5	29.0	27.0	25.0
40	6	—	—	51.0	48.5	46.0	43.5	41.0	38.5	36.0	33.5	31.0	28.5	—
45	6.5	58.5	58.0	53.5	51.0	48.5	45.5	43.0	40.5	37.5	35.0	32.5	30.0	—
50	7	61.0	58.5	55.5	53.0	50.5	47.5	45.0	42.0	39.0	36.5	34.0	—	—
55	7.5	63.5	60.5	58.0	55.0	52.0	49.5	46.5	43.5	40.5	38.0	—	—	—
60	8	65.5	62.5	59.5	57.0	54.0	51.0	48.0	45.0	42.0	39.0	—	—	—
65	8.5	67.5	64.5	61.5	58.5	55.5	52.5	49.5	46.5	43.5	40.5	—	—	—
70	9	69.0	66.5	63.5	60.5	57.0	54.0	51.0	48.0	45.0	41.5	—	—	—
75	9.5	71.0	68.0	65.0	62.0	59.0	55.5	52.5	49.0	46.0	—	—	—	—
80	10	72.5	69.5	66.5	63.5	60.0	57.0	54.0	50.5	47.0	—	—	—	—
85	10.5	74.5	71.5	68.0	65.0	61.5	58.5	55.0	51.5	48.0	—	—	—	—
90	11	76.0	73.0	69.5	66.5	63.0	59.5	56.0	53.0	49.0	—	—	—	—
95	11	77.5	74.5	71.0	67.5	64.5	61.0	57.0	54.0	—	—	—	—	—
100	11	79.0	76.0	72.5	69.0	65.5	62.0	58.5	55.0	—	—	—	—	—

(Continued)

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Table 13 (Concluded)

Class	H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7
Minimum circumference at top, in	**39**	**37**	**35**	**33**	**31**	**29**	**27**	**25**	**23**	**21**	**19**	**17**	**15**
Length of pole, ft	**Groundline distance from butt, ft***	**Minimum circumference at 6 ft from butt, in**											
105	12	80.5	77.0	74.0	70.5	67.0	63.0	59.5	56.0	—	—	—	—	—
110	12	82.0	78.5	75.0	71.5	68.0	64.5	60.5	57.0	—	—	—	—	—
115	12	83.5	80.0	76.5	72.5	69.0	65.5	61.5	58.0	—	—	—	—	—
120	12	85.0	81.0	77.5	74.0	70.0	66.5	62.5	59.0	—	—	—	—	—
125	12	86.0	82.5	78.5	75.0	71.0	67.5	63.5	59.5	—	—	—	—	—

* *These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

Note: *See Note 3) to Table 3 for the classification of coastal Douglas fir.*

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Table 14
Dimensions of eastern white cedar reinforcing stubs (fibre stress 3700 psi)
(See Clauses 4.1.2, 5.2, C.4, and E.1.)

Length of stub, ft	Minimum circumference at 6 ft from butt, in			
	Class Z	Class Y	Class X	Class W
10	44.5	38.5	33.0	28.0
11	51.0	44.0	37.5	32.0
13	56.0	49.0	41.5	—
15.5	58.5	51.0	—	—

Table 15
Dimensions of spruce and eastern white pine reinforcing stubs (fibre stress 4900 psi)
(See Clauses 4.1.2, 5.2, C.4, and E.1.)

Length of stub, ft	Minimum circumference at 6 ft from butt, in			
	Class Z	Class Y	Class X	Class W
10	40.5	35.0	30.0	25.5
11	46.0	40.0	34.0	29.0
13	51.0	44.0	37.5	—
15.5	53.0	46.0	39.5	—

Table 16
Dimensions of amabilis fir reinforcing stubs (fibre stress 5400 psi)
(See Clauses 4.1.2, 5.2, C.4, and E.1.)

Length of stub, ft	Minimum circumference at 6 ft from butt, in			
	Class Z	Class Y	Class X	Class W
10	39.0	34.0	29.0	24.5
11	44.5	38.5	33.0	—
13	49.0	42.5	36.5	—
15.5	51.0	44.5	—	—
19.5	56.5	49.0	—	—

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Table 17
Dimensions of western red cedar and ponderosa pine reinforcing stubs (fibre stress 5600 psi)
(See 4.1.2, 5.2, C.4, and E.1.)

Length of stub, ft	Minimum circumference at 6 ft from butt, in			
	Class Z	Class Y	Class X	Class W
10	38.5	33.5	28.5	24.0
11	44.0	38.0	32.5	—
13	48.5	42.0	36.0	—
15.5	50.0	44.0	—	—
19.5	56.0	48.5	—	—

Table 18
Dimensions of red pine reinforcing stubs (fibre stress 6000 psi)
(See Clauses 4.1.2, 5.2, C.4, and E.1.)

Length of stub, ft	Minimum circumference at 6 ft from butt, in			
	Class Z	Class Y	Class X	Class W
10	37.5	32.5	28.0	23.5
11	43.0	37.5	31.5	27.0
13	47.5	41.0	35.0	30.0
15.5	49.5	43.0	36.5	—

Table 19
Dimensions of Jack pine reinforcing stubs (fibre stress 6400 psi)
(See Clauses 4.1.2, 5.2, C.4, and E.1.)

Length of stub, ft	Minimum circumference at 6 ft from butt, in			
	Class Z	Class Y	Class X	Class W
10	37.0	32.0	27.0	23.0
11	42.0	36.5	31.0	—
13	46.5	40.0	34.5	—
15.5	48.0	42.0	—	—

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RsxjsvViwepi0$4<35<3645:$56:98=$Q HX

Table 20
Dimensions of yellow cedar and lodgepole pine reinforcing stubs (fibre stress 6600 psi)

(See Clauses 4.1.2, 5.2, C.4, and E.1.)

Length of stub, ft	Minimum circumference at 6 ft from butt, in			
	Class Z	Class Y	Class X	Class W
10	36.5	31.5	27.0	22.5
11	41.5	36.0	30.5	—
13	46.0	40.0	34.0	—
15.5	47.5	41.5	—	—
19.5	52.5	46.0	—	—

Table 21
Dimensions of western hemlock and interior Douglas fir reinforcing stubs (fibre stress 6800 psi)

(See Clauses 4.1.2, 5.2, C.4, and E.1.)

Length of stub, ft	Minimum circumference at 6 ft from butt, in			
	Class Z	Class Y	Class X	Class W
10	36.0	31.5	26.5	22.5
11	41.0	35.5	30.5	—
13	45.5	39.5	33.5	—
15.5	47.0	41.0	—	—
19.5	52.0	45.5	—	—

Table 22
Dimensions of southern yellow and Scots pine reinforcing stubs (fibre stress 7800 psi)

(See Clauses 4.1.2, 5.2, C.4, and E.1.)

Length of stub, ft	Minimum circumference at 6 ft from butt, in			
	Class Z	Class Y	Class X	Class W
10	34.5	30.0	25.5	21.5
11	39.0	34.0	29.0	24.5
13	43.0	37.5	32.0	27.0
15.5	45.0	39.0	33.5	—

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Table 23
Dimensions of western larch and coastal Douglas fir reinforcing stubs (fibre stress 8000 psi)

(See Clauses 4.1.2, 5.2, C.4, and E.1.)

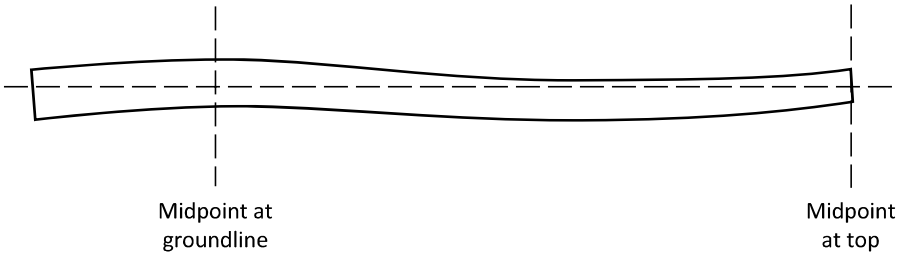
Length of stub, ft	Minimum circumference at 6 ft from butt, in			
	Class Z	Class Y	Class X	Class W
10	34.0	29.5	25.0	21.0
11	39.0	33.5	28.5	—
13	43.0	37.0	31.5	—
15.5	44.5	38.5	—	—
19.5	49.0	43.0	—	—

Note: *See Note 3) to Table 3 for the classification of coastal Douglas fir.*

①

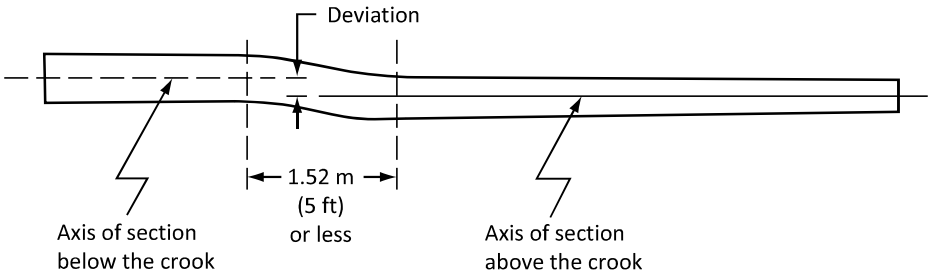
Figure 1
Measurement of sweep in poles

(See Clauses 3 and 4.4.12.)

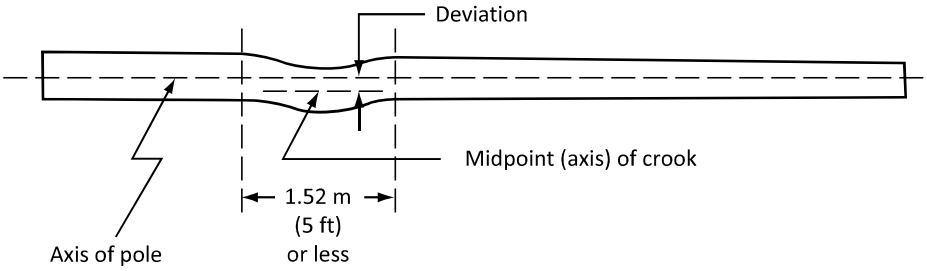


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RsxjsvViwepi0$4<35<3645:$56:98=$Q HX

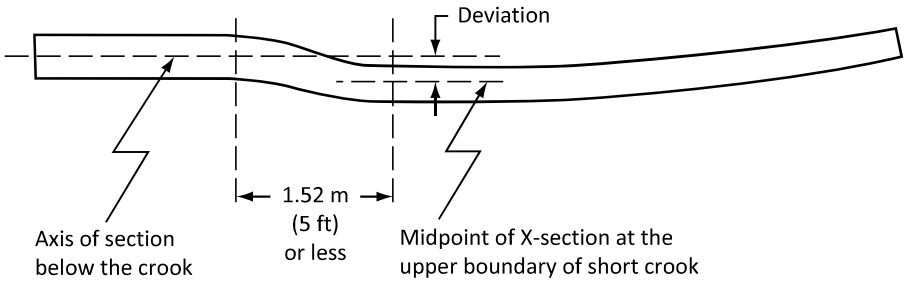
Figure 2
Measurement of short crook in poles
(See Clauses 3 and 4.4.12.)



a) Case 1: Reference axes are approximately parallel



b) Case 2: Axes of sections above and below the crook coincide or are practically coincident



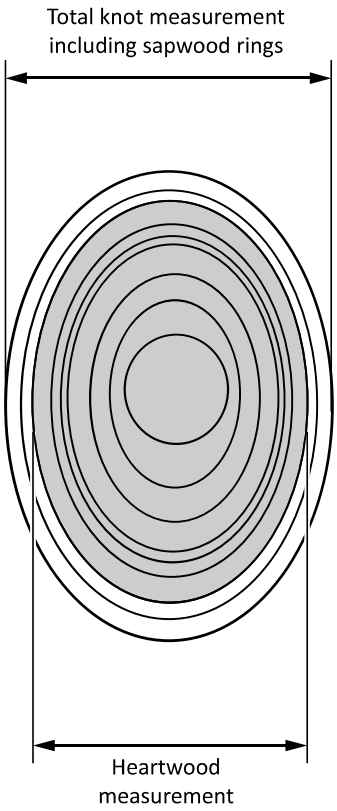
c) Case 3: Axis of the section above the short crook is not parallel to or coincident with the axis below the crook

Note: *The three cases shown in this Figure are typical and are intended to establish the principle of measuring short crooks. There can be other cases not exactly like those illustrated.*

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Gst}wklxGWEKvsyt$
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RsxjsvViwepi0$4<35<3645:$56:98=$Q HX

Figure 3
Knot measurement
(See Clause 4.4.7.)



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RsxjsvViwepi0$4<35<3645:$56:98=$Q HX

Annex A (informative)
Database for classification stresses

Notes:
1) *This Annex is not a mandatory part of this Standard.*
2) *The strength values specified in this Annex are expressed in yard/pound units, i.e., in the form in which they were originally derived.*

A.1 Objective
The objective of this Annex is to present a database for classification stresses (see Table A.1) considered to be representative of material that is available to the Canadian market.

A.2 Definitions
The following definitions apply in this Annex:

Note: *The following definitions provide a guide to the column headings in Table A.1. The abbreviations in parentheses following the defined terms reflect the abbreviations in the column headings. See also Note 4) to Table A.1.*

Length — the nominal length of poles in the sample. The testing span is smaller and generally in accordance with ASTM D1036.

Modulus of elasticity (MOE) — the average modulus of elasticity in 69 bar (10^3 lbs/in^2) units calculated, in most cases, in accordance with ASTM D1036.

Modulus of rupture (MOR) — the average modulus of rupture or maximum breaking stress in pounds per square inch.

Note: *Users should not infer that all poles broke at the groundline. Many broke at a smaller cross-section where the calculated extreme fibre stress was usually less, but might have been greater, than that at the groundline cross-section.*

Reference number (Ref. no.) — the number assigned in this Annex's bibliography (see Clause A.9) to the document on which the related information in Table A.1 is based.

Standard deviation — a statistical measure of variability.

S_1 — the standard deviation of the modulus of rupture found in the sample.

S_2 — the standard deviation of the modulus of elasticity found in the sample.

Site number (Site no.) — the site or pole yard from which the sample was obtained (see Table A.2). For some sites, additional identifying information can be found in the applicable document referenced in Clause A.9.

Treatment — the code (see Clause A.8) describing the conditioning and treatment of poles before strength testing.

Note: *Further information can be found in the applicable document referenced in Clause A.9.*

A.3 Strength determination
A reasonable estimate of the percentage of a sample of poles having strengths falling within a specific range can be obtained using normal probability theory. On this basis, the range expressed by ± 1

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Rs$vitvshyqxmsr$sv$rix{svomrk$tivqmxihs{mdsyx$pmgirwi$jsq$LW
Pmgirwi AQ ermsfe$L}hvs9=4<5==5440Wwi vA[s{vjo0Newsr
RsxjsvViwepi0$4<35<3645:$56:98=$Q HX

standard deviation from the average value will include about 68% of cases, ± 2 standard deviations from the average will include about 95% of cases, and ± 3 standard deviations from the average will include about 99.7% of cases. Thus, from a sample of poles having an average modulus of rupture of 552 bar (8000 psi) and a standard deviation of 69 bar (1000 psi), the strengths of 68% of the sample can be expected to fall within the range of 483 to 621 bar (7000 to 9000 psi), 95% within the range of 414 to 690 bar (6000 to 10 000 psi), and 99.7% within the range of 345 to 758 bar (5000 to 11 000 psi).

A.4 Coefficient of variation

If a coefficient of variation for any sample is desired, it may be calculated by expressing the appropriate standard deviation as a percentage of the average value.

A.5 Sample exclusions

Some samples in Table A.1 have been excluded from the weighted species averages in the rows labelled "Average" because they were considered extraordinary on account of length, kerfing, treatment, moisture content, or decay.

A.6 Moisture content

The moisture content of most specimens when tested was above the fibre saturation point, as required by ASTM D1036, which has been used since 1949 for the testing of utility poles. The fibre saturation point is the moisture content level at which swelling of the wood ceases or shrinkage begins. A moisture content above the fibre saturation point is a condition at the groundline recognized as the best approximation for the purposes of this Standard.

A.7 Additional data

Further information on the samples listed in Table A.1 and on the derivation of the numbers in that Table is archived at Forintek Canada Corp.

A.8 Seasoning and treatment numbers

A.8.1

"Nil" in the "Treatment" column in Table A.1 means no preservative treatment. Some of the samples were tested in the green condition or in a condition of moisture content greater than the fibre saturation point. Other samples were seasoned and then butt soaked before they were tested for bending strength.

Δ A.8.2

The treatment codes in Table A.1 consist of three digits, as follows:

a) The first digit refers to seasoning:
   i) 0 means no seasoning or unknown;
   ii) 1 means air drying;
   iii) 3 means Boultonizing or boiling under vacuum;
   iv) 4 means steaming; and
   v) 6 means kiln drying.

b) The second digit refers to the chemical used:
   i) 0 means no preservative treatment or unknown;
   ii) 1 means 100% creosote;
   iii) 3 means pentachlorophenol;

Gst}wklxGWEKvsyt$
Hsgyqirx$tvszmihs$}9LW$yrhivqmirwi$jsqGWEKvsyt
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RsxjsvViwepi0$4<35<3645:$56:98=$Q HX

iv) 5 means ACA; and
v) 8 means unknown.

c) The third digit refers to the method of treatment:
i) 0 means no method of treatment or unknown;
ii) 1 means Rueping process;
iii) 2 means Lowry process;
iv) 3 means butt treatment;
v) 4 means Bethel process; and
vi) 6 means thermal treatment.

A.9 References

The following references apply to the "Ref. no." column in Table A.1:

1) Alexander, J.B. 1934. *The strength of lodgepole pine telephone poles*. Western Forest Products Laboratory Circular 42. Vancouver.
2) *The strength of telephone poles of eastern cedar, red pine and Jack pine*. 1947. Eastern Forest Products Laboratory Circular 31 (revised). Ottawa.
3) Aplin, E.N. 1967. *The strength of red pine poles.* Eastern Forest Products Laboratory Report OP-X-IO. Ottawa.
4) Fox, S.P. 1977. Unpublished data on strength of ACA- and PCP-treated white spruce, western hemlock, and amabilis fir. Western Forest Products Laboratory. Vancouver.
5) McGowan, W.M. 1960. *The strength of Douglas fir telephone poles.* Western Forest Products Laboratory Technical Note No. 15. Vancouver.
6) McGowan, W.M. 1962. *The strength of western hemlock power and communication poles.* Western Forest Products Laboratory Technical Note No. 27. Vancouver.
7) McGowan, W.M., and W.J. Smith. 1965. *Strength and related properties of western red cedar poles.* Western Forest Products Laboratory Publication 1108. Vancouver.
8) Platt, J.C. 1958. *Strength tests of Jack pine poles.* Ontario Hydro Report 58-149.
9) Platt, J.C. 1958. *The strength of wood poles: Western red cedar and red pine.* Ontario Hydro Report 58-206.
10) Wood, L.W., E.G.O. Erickson, and A.W. Dohr. 1960. *Strength and related properties of wood poles (final report).* ASTM Wood Pole Research Program. Philadelphia: American Society for Testing and Materials.
11) Behr, E.A., B.M. Radcliffe, and K.O. Wilkins. 1965. Strength of plantation grown and second growth red pine poles. *Michigan agriculture export statistics*, *quarterly bulletin.* 48:296–319.
12) Kennedy, D.E., and W.E. Wakefield. 1948. *The strength of Jack pine poles infected with pocket rot.* Dominion Forest Service Circular 65. Ottawa: Department of Mines and Resources, Forest Products Laboratory.
13) Jessome, A.P. 1955. *An Investigation of the bending strength of 25 foot red pine poles by the machine test method.* Forest Products Laboratory Project 0-74-1, Progress Report 3. Ottawa: Department of Northern Affairs and Natural Resources.
14) Jessome, A.P. 1960. *An investigation of the bending strength of 35 foot untreated lodgepole pine poles by the machine test method.* Forest Products Laboratory Project 0-74-1, Progress Report 5. Ottawa: Department of Northern Affairs and Natural Resources.
15) Stewart, D.L. 1928. *Mechanical strength of red pine poles.* Bell Telephone.
16) Fox, S.P. Unpublished data by W.M. McGowan on the strength of new and used western red cedar poles.
17) McElhannay, T.A., and R.S. Perry. 1928. *Report on tests of the relative strength of green-cut and fire-killed western cedar pole timber.* Forest Service Circular No. 22. Ottawa. Department of the Interior.

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RsxjsvViwepi0$4<35<3645:$56:98:=$Q HX

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Note: *See the following for further information:*

a) *ASTM D1036; and*
b) *Jessome, A. P. 1977. Strength and related properties of woods grown in Canada. Eastern Forest Products Laboratory Technical Report 21. Ottawa.*

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Gst}wklxGWEKvsyt$
Hsgyqirx$tvszmih$F}WLWyrhivfpgirwi$jsvq$GWE$Kvsyt
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Pmgirwi AQ ermsfe$L}hvs9=4<5==5440$Wwiv A[s{vjo0$Newsr
RsxjsvViwepi0$4<35<3645:$56:98=$Q HX

Δ

Table A.1
Database for classification stresses
(See Clauses A.1, A.2, A.5, and A.7–A.9 and Tables 3, A.2, and E.3.)

Species	Site no. (see Table A.2)	Treatment (see Clause A.8)	No. of poles in sample	Strength values derived from references					
				Length, ft	MOR, psi	S_1, psi	MOE, 1000 psi	S_2, 1000 psi	Ref. no. (see Clause A.9)
Cedar, Eastern white	1	Nil	28	25	3658	656	—	—	2
Cedar, Western red	6	Nil	51	30	4587	599	980	126	7
	7	Nil	51	30	5229	763	1123	147	7
	6	Nil	40	30	5787	877	1148	148	7
	6	Nil	40	30	5620	827	1072	135	7
	25	Nil	35	30	5588	726	1223	133	10
	25	Nil	6	25	5322	501	1225	133	10
	29	003	25	30	4694	886	—	—	9
	25	136	35	30	5367	768	1214	128	10
	25	136, 036	5	30	4720	227	1146	81	10
	25	136	5	25	5750	884	1285	93	10
	38	Nil	15	25	7035	656	—	—	17
	39	Nil	12	25	6079	540	—	—	17
	40	Nil	9	25	6941	882	—	—	17
	38	Nil	34	25	6663	1009	—	—	17
	39	Nil	24	25	5850	962	—	—	17
Average*	—	—	—	—	5555	1054	1123	159	—
	25	Nil	5	55	3944	1070	1174	118	10

(Continued)

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Table A.1 (Continued)

Species	Site no. (see Table A.2)	Treatment (see Clause A.8)	No. of poles in sample	Strength values derived from references					
				Length, ft	MOR, psi	S_1, psi	MOE, 1000 psi	S_2, 1000 psi	Ref. no. (see Clause A.9)
	25	136	5	55	4318	557	1101	86	10
	32	Nil	50	35	6029	954	1147	197	16
	32	032	50	35	5340	909	1089	133	16
	7	Nil	70	35	4150	1071	870	225	16
Fir, Amabilis	32	654	51	35	5194	781	1161	247	4
	32	634	52	35	5642	1032	1262	206	4
Average	—	—	—	—	5420	963	1212	236	—
Fir, Coastal Douglas	11	Nil	28	25	8030	849	—	—	5
	11	111	27	25	8242	1185	—	—	5
	12	111	24	25	9133	1376	—	—	5
	21	Nil	35	30	8185	948	2100	268	10
	21	Nil	5	25	8070	385	2093	139	10
	21	311	39	30	7718	837	2045	220	10
	21	311	5	25	7596	1145	2104	239	10
Average (Coastal fir)	—	—	—	—	8174	1021	2074	239	—
Fir, Interior Douglas	13	Nil	50	25	6858	1096	—	—	5
	13	111	49	25	6751	938	—	—	5
	†	Nil	6	35	6928	1478	—	—	29
	†	Nil	2	40	7431	710	—	—	29
	†	Nil	4	45	7173	709	—	—	29
	†	Nil	6	50	6797	1109	—	—	29
Average (Interior fir)	—	—	—	—	6834	1038	—	—	—

(Continued)

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Table A.1 (Continued)

Species	Site no. (see Table A.2)	Treatment (see Clause A.8)	No. of poles in sample	Strength values derived from references					
				Length, ft	MOR, psi	S_1, psi	MOE, 1000 psi	S_2, 1000 psi	Ref. no. (see Clause A.9)
Fir, Douglas (U.S.)	21	Nil	35	30	8185	948	2100	268	10
	21	Nil	5	25	8070	385	2093	139	10
	21	311	39	30	7718	837	2045	220	10
	21	311	5	25	7596	1145	2104	239	10
Average (U.S.)	—	—	—	—	7926	913	2074	239	—
Fir, Douglas (Coastal, Interior, and U.S.)									
Species average*	—	—	—	—	7657	1289	2074	239	—
	21	Nil	5	55	8226	605	2200	108	10
	21	311	5	55	6994	548	1632	88	10
Hemlock, Western	5	110	50	25	6432	975	1643	217	6
	5	Nil	52	25	6135	739	1662	256	6
	33	654	50	35	6967	1034	1365	216	4
	33	634	52	35	7483	1026	1430	186	4
Average	—	—	—	—	6755	1118	1525	268	—
Larch, Western	22	Nil	43	30	8551	998	2058	291	10
	22	Nil	6	25	8983	720	2005	177	10
	22	311	40	30	7071	1002	1649	249	10
	22	311	5	25	7726	1221	1901	213	10
Average*	—	—	—	—	7905	1368	1873	353	—
	22	Nil	5	55	7478	509	1844	144	10
	22	311	5	55	6912	1143	1379	255	10

(Continued)

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Table A.1 (Continued)

Species	Site no. (see Table A.2)	Treatment (see Clause A.8)	No. of poles in sample	Strength values derived from references					
				Length, ft	MOR, psi	S_1, psi	MOE, 1000 psi	S_2, 1000 psi	Ref. no. (see Clause A.9)
Pine, Jack	2	Nil	22	25	7217	605	—	—	2
	3	Nil	25	25	7860	634	—	—	2
	16	181	25	25	5241	664	—	—	8‡
	17	181	25	25	5840	954	—	—	8‡
	18	181	25	25	6183	997	—	—	8‡
	18	Nil	25	25	6096	821	—	—	8
	18	Nil	25	25	6070	832	—	—	8
	36	Nil	18	25	8304	752	—	—	12
Average*	—	—	—	—	6529	1296	—	—	—
	36	Nil	82	25	8012	1070	—	—	12
	36	Nil	26	25	7136	1086	—	—	12
Pine, Lodgepole (Canada)	8	Nil	23	25	5809	627	—	—	1
	8	111	24	25	6671	863	—	—	1
	9	Nil	23	25	6604	667	—	—	1
	10	Nil	25	25	6581	527	—	—	1
	37	Nil	15	35	6677	701	1636	181	14
	37	Nil	15	35	7259	1012	—	—	19
Average (Canada)*	—	—	—	—	6553	842	1636	181	—
Pine, Lodgepole (U.S.)	23	Nil	34	30	4878	851	1183	186	10
	23	Nil	6	25	4955	542	1175	142	10
	23	111	40	30	5085	786	1219	163	10
	23	111	5	25	5174	492	1120	120	10
Average (U.S.)	—	—	—	—	4998	782	1196	169	—

(Continued)

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Table A.1 (Continued)

Species	Site no. (see Table A.2)	Treatment (see Clause A.8)	No. of poles in sample	Strength values derived from references					
				Length, ft	MOR, psi	S_1, psi	MOE, 1000 psi	S_2, 1000 psi	Ref. no. (see Clause A.9)
Pine, Lodgepole (Canada and U.S.)									
Species average*	—	—	—	—	5924	1151	1262	251	—
	23	Nil	5	45	4220	648	1218	187	10
	23	111	5	45	4764	480	1060	88	10
Pine, Red	2	Nil	27	25	6761	1060	—	—	2
	4	Nil	25	25	6500	639	—	—	2
	2	110	25	25	6451	611	—	—	2
	14, 27, 28	Nil	15	30	5284	909	1372	196	3
	15	431	28	30	5434	807	1297	176	3
	30	031	25	30	5749	657	—	—	9
	41	111	6	25	5290	747	—	—	15
	2	112	3	25	6963	391	—	—	15
	41	111	6	30	5530	688	—	—	15
	2	112	3	30	7587	598	—	—	15
	41	111	6	35	5548	509	—	—	15
	2	112	3	35	7910	149	—	—	15
	50	Nil	6	35	4953	529	1143	157	28
	50	Nil	13	30	4910	531	1095	189	28
	51	Nil	3	35	6086	750	1272	135	28
	51	Nil	14	30	6231	487	1252	91	28
Average*	—	—	—	—	5990	1002	1257	187	—
	34	Nil	32	30	4802	501	842	98	11
	34	Nil	32	30	4916	414	873	121	11

(Continued)

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Table A.1 (Continued)

Species	Site no. (see Table A.2)	Treatment (see Clause A.8)	No. of poles in sample	Strength values derived from references					
				Length, ft	MOR, psi	S_1, psi	MOE, 1000 psi	S_2, 1000 psi	Ref. no. (see Clause A.9)
	35	Nil	25	25	5344	819	—	—	13
	35	Nil	25	25	5122	988	—	—	18
	41	100	6	25	5407	426	—	—	15‡
	41	100	11	30	6072	593	—	—	15‡
	41	100	6	35	5470	615	—	—	15‡
	41	100	3	40	5340	—	—	—	15‡
	50	Nil	1	35	4633	—	1073	—	28
	50	Nil	13	30	4515	529	956	75	28
Pine, Scots	42	110	10	—	6530	991	1200	186	21
	42	110	20	22	8148	917	1635	185	22
	43	110	10	—	8160	813	1490	130	21
	45	110	10	22	8510	1126	1750	277	22
	46	110	10	—	8220	615	1550	96	21
	47	110	10	—	7210	916	1340	193	21
Average*	—	—	—	—	7846	1132	1514	265	—
	44	Nil	12	20	9028	694	1950	400	23
	49	110	30	30	6730	—	1803	—	24‡
	49	000	5	—	6251	—	—	—	25‡
	45	080	16	—	7600	—	—	—	26
	48	Nil	5	—	8210	—	—	—	27
	48	110	5	—	8590	—	—	—	27‡

(Continued)

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Table A.1 (Concluded)

Species	Site no. (see Table A.2)	Treatment (see Clause A.8)	No. of poles in sample	Strength values derived from references					
				Length, ft	MOR, psi	S_1, psi	MOE, 1000 psi	S_2, 1000 psi	Ref. no. (see Clause A.9)
Pine, Southern yellow§	19, 20	Nil	20	25	9543	994	2123**	150**	10
	19, 20	Nil	70	30	8752	1196	1885††	285††	10
	31	411	50	30	5747	974	1173	178	20
	31	611	50	30	7902	1018	1272	184	20
Average*	—	—	—	—	7821	1897	1507‡‡	471‡‡	—
	19, 20	Nil	20	55	8129	1192	1724	208	10
Spruce, White	26	134	53	35	4961	951	1061	192	4
	26	154	53	35	4911	685	986	134	4
Average*	—	—	—	—	4936	825	1023	178	—
	26	154	53	35	4420	680	902	157	4

* *The samples listed below the "Average" row for a particular species are those excluded from the calculation of the average. See Clause A.5for further information.*
† *Information not available.*
‡ *The moisture content of many poles in this sample was less than the fibre saturation point for the species.*
§ *Steam conditioning of poles can reduce bending strength. Southern yellow pine poles or other species used in Canada should be treated in accordance with the CAN/CSA-O80 Series of Standards.*
** *Based on ten specimens.*
†† *Based on 59 specimens.*
‡‡ *Based on 169 specimens.*

Notes:

1) *This database contains Canadian, U.S., and European data determined by institutions considered to be without bias; the test reports are held by Forintek Canada Corp. (Western Laboratory).*

2) *See Clauses A.2, A.5, A.8, and A.9 and Table A.2 for explanations of the material appearing in this Table.*

3) *The following conversion factors may be used with this Table:*

a) *1 MPa = 145.038 psi; and*

b) *1 m = 3.28084 ft.*

4) *MOR is the average modulus of rupture at the groundline cross-section in pounds per square inch or, if that value is not available, the average extreme fibre stress at the failing cross-section. S1 is the standard deviation of the MOR from which the coefficient of variation equal to S2 (100)/MOR may be calculated. MOE is the average modulus of elasticity in 103 pounds per square inch. S2 is the standard deviation of the MOE from which the coefficient of variation equal to 32 (100)/MOE may be calculated.*

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Table A.2
Site numbers for Table A.1
(See Clause A.2 and Table A.1.)

Site number	Location
	Ontario and adjacent areas
1	North Bay
3	Timmins
4	Fort Coulonge, Québec
14	Uxbridge
15	Pembroke
16–18	Three districts about 300 miles apart
27	Vivian
28	St. Williams
30	Eastern Ontario
34	Michigan
35	Eganville
36	Armstrong
41	Ottawa valley
50	Northern New York state (plantation)
51	Northern New York state (second growth)
	Coast of British Columbia and Oregon
5	Burrard Inlet
7	Southwest coast
11	Port Moody
12	Cowichan
21	Willamette National Forest, Oregon
32	Vancouver Island
33	Kelsey Bay
39	New Westminster
40	Harrison Lake
	Interior of British Columbia and adjacent areas
6	Lumby
8	Kingsgate
9	Rocky Mountain House, Alberta

(Continued)

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Table A.2 (Concluded)

Site number	Location
10	Crowsnest
13	Merritt
22	Northwestern Montana
23	Northern Colorado and southern Montana
25	Priest Lake area, Washington and Idaho
26	Dawson Creek
29	Unknown
37	Prince George
38	Shuswap Lake
	New Brunswick
2	Newcastle
	Southeastern United States
19	Lower coastal plain, southern South Carolina
20	Kisatchie National Forest, central Louisiana
31	Unknown
	Europe
42	New Forest, England
43	Nairn, Scotland
44	Windsor Forest, England
45	Sweden
46	Finland
47	Norway
48	Europe
49	Unknown

Gst}wklxGWEKvsyt$
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RsxjsvViwepi0$4<35<3645:$56:98=$QHX

Annex B (informative)
Classification procedure for poles

Note: *This Annex is not a mandatory part of this Standard.*

B.1 Derivation

B.1.1

The minimum circumferences specified in Tables 4 to 13 have been calculated in order for each species in a given class to develop groundline stresses approximately equal to those specified in Table 3 when a given horizontal load is applied 0.61 m (2 ft) from the top of the pole. The horizontal loads used in the calculations for separating the thirteen classes are specified in Table B.1.

B.1.2

In making the calculations, it was assumed that the pole is used as a simple cantilever and that the maximum fibre stress in the pole subjected to the bending moment applied will occur at the groundline. The circumference at the groundline was calculated using the standard engineering formula. This value was then used to calculate the required circumference 1.83 m (6 ft) from the butt using a calculated taper per foot of length between the groundline and 1.83 m (6 ft) from the butt. The calculated circumference 1.83 m (6 ft) from the butt was rounded to the nearest 13 mm (1/2 in).

Note: *A calculated taper is the difference between the minimum top circumference and the groundline circumference divided by the length between them.*

B.1.3

The assumption that the maximum fibre stress in the pole occurs at the groundline is theoretically correct, provided that the circumference at the groundline is not more than 1-1/2 times the circumference at the point of loading. If the circumference at the groundline is more than 1-1/2 times the circumference at the point of loading, theoretically the maximum fibre stress in the pole occurs above the groundline at the point where the circumference is 1-1/2 times that at the point of loading (not necessarily the circumference 0.61 m (2 ft) from the top). When the load is not applied 0.61 m (2 ft) from the top, it is necessary for the class of pole to be determined by making separate calculations from the specific loads to be supported.

Unless other information is available to substantiate the actual taper of the pole for a given species, these calculations should be based on the taper determined by the difference in circumference between the specified top dimension and the groundline dimension divided by the distance in feet between the two measurements.

B.2 Shrinkage

The dimensions specified in Tables 4 to 13 are the minimum dimensions of the poles when offered to the purchaser, regardless of the state of seasoning. While it is recognized that some shrinkage will occur in seasoning, it is known that the dimensional loss is offset by an increase in fibre strength.

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Gst}wklxGWEKvsyt$
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Table B.1
Horizontal loads
(See Clause B.1.1 and Tables 7 –9 and E.7 –E.9.)

Class	Horizontal load	
	N	lbf
H6	50 700	11 400
H5	44 500	10 000
H4	38 700	8 700
H3	33 400	7 500
H2	28 500	6 400
H1	24 000	5 400
1	20 000	4 500
2	16 500	3 700
3	13 300	3 000
4	10 700	2 400
5	8 500	1 900
6	6 700	1 500
7	5 300	1 200
8	4 300	960

Note: *The loads specified in this Table should be corrected by a factor of 0.95 for western red cedar, red pine, and Jack pine.*

Gst}wklxGWEKvsyt$
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Pmgirwi AQ ermsfe$L}hvs9=4<5==5440$Wiv A[s{vjo0$Newsr
RsxjsvViwepi0$4<35<3645:$56:98=$Q HX

Annex C (informative)
Classification procedure for reinforcing stubs

Note: *This Annex is not a mandatory part of this Standard.*

C.1 Classification

Four classes of reinforcing stubs were adopted for use with Class 1 to Class 8 poles. The adopted classes of reinforcing stubs and associated classes of poles are as follows:

Class of stub	Class of associated poles
Z	1 and 2
Y	3 and 4
X	5 and 6
W	7 and 8

C.2 Lengths

Five lengths of reinforcing stubs were adopted, as follows:

Length of stub, ft	Length of associated poles, ft
10	20, 25, 30
11	35, 40
13	45, 50
15.5	55
19.5	60, 70

C.3 Size determination

It is recognized that a pole will be stub reinforced usually after it has served a substantial part of its physical life. Consequently, a stub should have a life at least equal to the remaining life of the pole. Experience has indicated that this result will generally be met by selecting a reinforcing stub whose circumference at 1.83 m (6 ft) from the butt is approximately 51 mm (2 in) less than the circumference at 1.83 m (6 ft) from the butt of a new replacement pole adequate to support the load. From these selected bases, the dimensions of new stubs have been determined, the length and the groundline circumference being designed to develop the strength of the longest pole of the highest class with which the stub is intended to be associated. For example, a 3.05 m (10 ft) Class Z stub is designed to develop the expected remaining strength of a 9.1 (30 ft) Class 1 pole.

11dd0ddddddd000d000d0dd0000ddddd01d1d00d00d0d00d111
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Pmirwi AQ ermsfe$L}hvs9=4<5==5440Wwiv A[s{vjo0Newsr
RsxjsvViwepi0$4<35<3645:$56:98=$Q HX

C.4 Shrinkage

The dimensions specified in Tables 14 to 23 are the minimum dimensions of the poles when offered to the purchaser, regardless of the state of seasoning. While it is recognized that some shrinkage will occur in seasoning, it is known that the dimensional loss is offset by an increase in fibre strength.

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11dd0ddddddd0000d000d0dd0000ddddd01d1d000d00d0d0d111

Gst}vklxGWEKvsyt$
Hsgyq irx$tvszmih$f}LWyrhiv$pgirwi$jsqGWEKvsyt
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RsxjsvViwepi0$4<35<3645:$56:98=$Q HX

Annex D (informative)
Ordering information

Note: *This Annex is not a mandatory part of this Standard.*

D.1

Poles and reinforcing stubs should be ordered by the piece and the following information should be provided:

a) quantity;
b) species;
c) length;
d) class;
e) preservative;
f) net retention of the preservative and CSA Standards or other references; and
g) roofing, gaining, and boring.

Note: *A sample order is as follows: quantity, species, length, and class; poles classified in accordance with this Standard and treated with (kg/m³) preservative in accordance with the CAN/CSA-O80 Series of Standards; details of roofing, gaining, and boring.*

D.2

Complete and detailed instructions should be given to the supplier whenever the requirements of this Standard are modified to meet special conditions.

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Gst}vklxGWEKvsyt$
Hsgyqirx$tvszmihs$}LWyrhivfpgirwijsqGWE$Kvsyt
Rs$vitvshyqpmsr$sv$rix{svcmrk$tivqmxihs${mdsyx$pgirwijsqLW
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RsxjsvViwepi0$4<35<3645:$56:98=$Q HX

Annex E (informative)
SI (metric) versions of Tables 1 to 23

Note: *This informative Annex has been written in mandatory language for users working with SI units. Users of this Standard should note that pole lengths given in SI units differ slightly in length from their imperial counterparts.*

Δ E.1

Tables E.1 to E.23 provide SI (metric) equivalents for Tables 1 to 23.

The dimensions specified in Tables 4 to 23 have been converted to SI units in Tables E.4 to E.23 by recalculating the theoretical circumference at 1.83 m (6 ft) from the butt in yard/pound units. The results have then been converted to centimetres and rounded to the nearest centimetre. This is thought to be more accurate than converting the values in Tables 4 to 23, which are rounded to the nearest 13 mm (1/2 in).

In cases of dispute, the dimensions specified in Tables 1 to 23 should govern.

Table E.1
Maximum knot sizes
(See Clause E.1.)

	Maximum knot sizes		
Length of pole, m	**Diameter of any single knot, mm**		**Sum of diameters of all knots in any length of 0.3 m**
	Classes H6 to 3	Classes 4 to 7	All classes
13.7 and shorter	102	76	One-third of the average circumference of the same 0.3 m section or 203 mm, whichever is greater, but not to exceed 31 cm
15.2 and longer	127	127	One-third of the average circumference of the same 0.3 m section or 254 mm, whichever is greater, but not to exceed 36 cm

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RsxjsvViwepi0$4<35<3645:$56:98=$Q HX

Table E.2
Maximum spiral grain
(See Clause E.1.)

Length of pole, m	Maximum twist of grain permitted
9.1 and shorter	1 complete twist in any 3 m length
10.7–13.7	1 complete twist in any 4.9 m length
15.2 and longer	1 complete twist in any 6.1 m length

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Table E.3
Strength and stiffness properties by species
(See Clause E.1 and Tables E.5, E.7, E.10, E.11, 13, and E.23.)

Common name	Botanical name	Fibre stress, MPa	Fibre stress coefficient of variation, %	Modulus of elasticity, MPa	Modulus of elasticity coefficient of variation, %
Cedar					
Eastern white	*Thuja occidentalis*	26	18	—	—
Western red	*Thuja plicata*	39	19	7720	14
Yellow	*Chamaecyparis nootkatensis*	46	—	—	—
Fir					
Douglas	*Pseudotsuga menziesii*				
Coastal	—	55	15	14 000	11
Interior	—	47	15	—	—
Amabilis	*Abies amabilis*	37	18	8810	19
Hemlock					
Western	*Tsuga heterophylla*	47	17	9580	15
Larch					
Western	*Larix occidentalis*	55	17	13 000	20
Pine					
Eastern white	*Pinus strobus*	34	—	—	—
Jack	*Pinus banksiana*	44	20	—	—
Lodgepole	*Pinus contorta*	46	13	8200	14
Ponderosa	*Pinus ponderosa*	39	—	—	—

(Continued)

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Table E.3 (Continued)

Common name	Botanical name	Fibre stress, MPa	Fibre stress coefficient of variation, %	Modulus of elasticity, MPa	Modulus of elasticity coefficient of variation, %
Red	*Pinus resinosa*	41	17	8890	14
Scots	*Pinus sylvestris*	54	14	10 480	18
Southern yellow	—	54	24	11 000	29
Spruce	—	34	17	7100	17

Notes:

1) *The fibre stresses are stresses for the purposes of classification and are close approximations to the average bending strengths (MOR) specified in Table A.1 for poles 6.1 to 10.7 m long.*

2) *The bending strengths specified in Table A.1 are derived from specimens whose moisture contents at the groundline were above the fibre saturation point at the time of testing to failure. This condition is recognized as the best approximation for the purposes of this Standard. Furthermore, the average moduli of elasticity specified in this Table are derived from poles of any length that were unseasoned or completely submerged in water for some time before testing, as described in the publications referenced in Clause A.9.*

3) *The strength of poles longer than 15.3 m can be less, as indicated in Annex A.*

4) *Fibre stress for several species was based on limited test information or results from one test site; therefore, the Subcommittee is uncertain of the adequacy of the pole dimensions for these species and recommends that further tests be performed to supplement the database. Notes concerning fibre stress accompany Tables 3 and E.3.*

5) *Other criteria related to the classification of poles are as follows:*

a) *The taper for determining the circumference at the classification point should be the difference between the minimum top circumference and the circumference at the theoretical groundline, divided by the length between them.*

b) *The loading point should be 0.61 m (2 ft) from the pole top.*

c) *The classification point should be 1.83 m (6 ft) from the butt.*

d) *The term "class" as used in this Standard does not refer to quality limitations or defects of a pole or reinforcing stub; it is used to differentiate poles and stubs on the basis of their potential force-resisting capabilities.*

6) *Pole test data for yellow cedar, eastern white pine, and ponderosa pine were not available; hence, the values specified in this Table approximate the average strengths of small, clear bending specimens, as described in the publications referenced in Clause A.9.*

7) *Samples of Douglas fir poles taken from coastal areas of British Columbia and from the Cascade Mountains region of Oregon have significantly higher average strengths than those taken from Rocky Mountain or interior areas. Thus, only Douglas fir poles taken from the western slopes of the Coast and Cascade Mountains (as defined by the Atlas of Canada and the National Atlas of the United States) shall be classified as coastal Douglas fir. All other Douglas fir poles shall be classified as interior Douglas fir.*

(Continued)

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Table E.3 (Concluded)

8) *In addition to the natural variability within a sample of poles, there is evidence from the data in Annex A that there is significant variability between pole samples taken from various areas of its growth region.*

9) *Scots and southern yellow pines are not native to Canada. Southern yellow pine is the name for a commercial grouping of species that includes loblolly (Pinus taeda), longleaf (P. palustris), shortleaf (P. echinata), and slash (P. elliottii) pines.*

10) *Spruce species include black (Picea mariana), Engelmann (P. engelmannii), red (P. rubens), Sitka (P. sitchensis), and white (P. glauca) spruce, of which only the latter has been tested in a pole size. Separation of red and black spruce from white spruce is impractical once the pole is removed from the forest. Small, clear bending specimens tested for each species indicate that white spruce is the weakest of the five.*

Table E.4
Dimensions of eastern white cedar poles (fibre stress 26 MPa)
(See Clause E.1.)

Class		1	2	3	4	5	6	7
Minimum circumference at top, cm		69	64	58	53	48	43	38
Length of pole, m	Groundline distance from butt, m*	Minimum circumference at 1.8 m from butt, cm						
6.1	1.2	98	91	85	79	73	67	62
7.6	1.5	109	102	95	88	81	75	70
9.1	1.7	118	111	103	96	89	82	76
10.7	1.8	127	119	111	103	95	88	82
12.2	1.8	134	126	117	109	101	93	86
13.7	2.0	141	132	123	114	106	98	—
15.2	2.1	148	138	129	120	111	—	—
16.8	2.3	154	144	134	125	—	—	—
18.3	2.4	159	149	139	129	—	—	—
19.8	2.6	164	154	144	133	—	—	—

** These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

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Table E.5
Dimensions of spruce and eastern white pine poles (fibre stress 34 MPa)
(See Clause E.1.)

Class		1	2	3	4	5	6	7
Minimum circumference at top, cm		69	64	58	53	48	43	38
Length of pole, m	Groundline distance from butt, m*	Minimum circumference at 1.8 m from butt, cm						
6.1	1.2	90	84	78	72	67	62	57
7.6	1.5	99	93	87	80	74	69	64
9.1	1.7	108	101	94	87	81	75	69
10.7	1.8	115	108	101	94	87	80	74
12.2	1.8	122	114	107	99	92	85	79
13.7	2.0	128	120	112	104	96	89	83
15.2	2.1	134	126	117	109	101	93	—
16.8	2.3	140	131	122	113	105	97	—
18.3	2.4	145	136	126	117	—	—	—
19.8	2.6	149	140	131	121	—	—	—

* *These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

Note: *See Note 2) to TableE.3 on eastern white pine*

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Table E.6
Dimensions of amabilis fir poles (fibre stress 37 MPa)
(See Clause E.1.)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7
Minimum circumference at top, cm		**99**	**94**	**89**	**84**	**79**	**74**	**69**	**64**	**58**	**53**	**48**	**43**	**38**
Length of pole, m	**Groundline distance from butt, m***	**Minimum circumference at 1.8 m from butt, cm**												
6.1	1.2	—	—	—	—	—	—	87	82	76	70	65	60	55
7.6	1.5	—	—	—	—	—	—	96	90	84	78	72	66	62
9.1	1.7	—	—	—	—	—	—	104	98	91	85	78	72	67
10.7	1.8	—	—	—	—	126	119	112	105	98	91	84	78	72
12.2	1.8	—	—	147	140	133	126	118	111	103	96	89	82	—
13.7	2.0	169	162	155	147	140	132	124	117	109	101	93	86	—
15.2	2.1	177	169	163	154	146	138	130	122	114	105	98	—	—
16.8	2.3	184	176	168	160	152	144	135	127	118	110	—	—	—
18.3	2.4	191	182	174	166	157	149	140	131	122	114	—	—	—
19.8	2.6	197	188	180	171	162	154	145	135	126	117	—	—	—
21.3	2.6	203	194	185	176	167	158	148	139	130	121	—	—	—
22.9	2.7	208	199	190	181	172	162	153	143	134	—	—	—	—
24.4	2.9	214	204	195	186	176	167	157	147	137	—	—	—	—
25.9	3.0	219	209	200	190	181	171	161	151	140	—	—	—	—
27.4	3.2	224	214	204	195	185	174	164	154	144	—	—	—	—
29.0	3.4	228	218	208	198	188	178	167	157	—	—	—	—	—

(Continued)

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Table E.6 (Concluded)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7
Minimum circumference at top, cm		99	94	89	84	79	74	69	64	58	53	48	43	38
Length of pole, m	Groundline distance from butt, m*	Minimum circumference at 1.8 m from butt, cm												
30.5	3.4	232	222	212	202	192	181	171	160	—	—	—	—	—
32.0	3.4	237	227	217	206	196	185	174	163	—	—	—	—	—
33.5	3.7	241	231	220	210	199	188	177	166	—	—	—	—	—
35.1	3.7	245	234	224	213	202	191	180	168	—	—	—	—	—
36.6	3.7	248	238	227	216	205	194	182	171	—	—	—	—	—
38.1	3.7	252	241	230	219	208	197	185	173	—	—	—	—	—

** These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

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Table E.7
Dimensions of western red cedar and ponderosa pine poles (fibre stress 39 MPa)
(See Clause E.1.)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7
Minimum circumference at top, cm		**99**	**94**	**89**	**84**	**79**	**74**	**69**	**64**	**58**	**53**	**48**	**43**	**38**
Length of pole, m	**Groundline distance from butt, m***	**Minimum circumference at 1.8 m from butt, cm**												
6.1	1.2	—	—	—	—	—	—	85	80	75	69	64	59	55
7.6	1.5	—	—	—	—	—	—	94	88	83	76	71	65	61
9.1	1.7	—	—	—	—	—	—	102	96	89	83	76	71	66
10.7	1.8	—	—	—	—	122	116	108	102	96	88	82	76	70
12.2	1.8	—	—	144	136	130	122	115	108	101	93	87	80	—
13.7	2.0	164	158	150	142	136	129	121	113	106	98	92	84	—
15.2	2.1	170	164	156	149	141	134	126	118	111	102	96	—	—
16.8	2.3	178	170	163	155	146	139	131	123	115	107	—	—	—
18.3	2.4	183	176	168	160	151	144	136	127	118	111	—	—	—
19.8	2.6	189	182	173	165	156	149	140	131	122	115	—	—	—
21.3	2.6	195	187	178	170	162	153	144	135	126	117	—	—	—
22.9	2.7	200	192	183	174	165	156	148	139	130	—	—	—	—
24.4	2.9	205	196	188	179	170	160	151	142	132	—	—	—	—
25.9	3.0	210	201	192	183	174	164	155	145	136	—	—	—	—
27.4	3.2	215	206	196	187	178	168	159	149	139	—	—	—	—
29.0	3.4	219	210	201	191	182	172	162	151	—	—	—	—	—

(Continued)

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Table E.7 (Concluded)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7
Minimum circumference at top, cm		**99**	**94**	**89**	**84**	**79**	**74**	**69**	**64**	**58**	**53**	**48**	**43**	**38**
Length of pole, m	**Groundline distance from butt, m***	**Minimum circumference at 1.8 m from butt, cm**												
30.5	3.4	223	214	205	195	184	173	165	155	—	—	—	—	—
32.0	3.4	228	217	209	198	188	178	168	158	—	—	—	—	—
33.5	3.7	231	221	212	202	192	182	172	160	—	—	—	—	—
35.1	3.7	235	225	215	205	195	184	174	163	—	—	—	—	—
36.6	3.7	239	229	219	209	198	188	177	165	—	—	—	—	—
38.1	3.7	242	233	223	211	201	191	179	168	—	—	—	—	—

** These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

Notes:

1) *For marketing purposes, yellow cedar can be considered the same as western red cedar on a dimensional basis only, but such poles shall be identified by actual species.*

2) *See Note 2) to Table E.3 on Ponderosa pine.*

3) *The loads specified in Table B.1 should be corrected by a factor of 0.95 for western red cedar.*

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Table E.8
Dimensions of red pine poles (fibre stress 41 MPa)
(See Clause E.1.)

Class		1	2	3	4	5	6	7	8
Minimum circumference at top, cm		**69**	**64**	**58**	**53**	**48**	**43**	**38**	**38**
Length of pole, m	**Groundline distance from butt, m***	**Minimum circumference at 1.8 m from butt, cm**							
6.1	1.2	83	78	73	68	62	57	54	51
7.6	1.5	92	85	79	74	69	64	59	56
9.1	1.7	99	93	87	80	74	69	64	61
10.7	1.8	106	98	92	85	79	73	68	65
12.2	1.8	112	104	97	90	84	78	—	—
13.7	2.0	117	109	102	94	88	82	—	—
15.2	2.1	122	115	107	99	92	—	—	—
16.8	2.3	126	118	111	103	—	—	—	—
18.3	2.4	131	122	115	107	—	—	—	—
19.8	2.6	135	126	117	109	—	—	—	—

** These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

Notes:

1) *There is evidence to suggest that red pine poles of average strength lower than 41 MPa can be obtained from certain areas of the United States.*

2) *The loads specified in Table B.1 should be corrected by a factor of 0.95 for red pine.*

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Table E.9
Dimensions of Jack pine poles (fibre stress 44 MPa)
(See Clause E.1.)

Class		1	2	3	4	5	6	7	8
Minimum circumference at top, cm		69	64	58	53	48	43	38	38
Length of pole, m	Groundline distance from butt, m*	Minimum circumference at 1.8 m from butt, cm							
6.1	1.2	83	78	73	68	62	57	54	51
7.6	1.5	92	85	79	74	69	64	59	56
9.1	1.7	99	93	87	80	74	69	64	61
10.7	1.8	106	98	92	85	79	73	68	65
12.2	1.8	112	104	97	90	84	78	—	—
13.7	2.0	117	109	102	94	88	82	—	—
15.2	2.1	122	115	107	99	92	—	—	—
16.8	2.3	126	118	111	103	—	—	—	—
18.3	2.4	131	122	115	107	—	—	—	—
19.8	2.6	135	126	117	109	—	—	—	—

* *These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

Note: *The loads specified in Table B.1 should be corrected by a factor of 0.95 for Jack pine.*

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Table E.10
Dimensions of yellow cedar and lodgepole pine poles (fibre stress 46 MPa)
(See Clause E.1.)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7	8
Minimum circumference at top, cm		**99**	**94**	**89**	**84**	**79**	**74**	**69**	**64**	**58**	**53**	**48**	**43**	**38**	**38**
Length of pole, m	**Groundline distance from butt, m ***	**Minimum circumference at 1.8 m from butt, cm**													
6.1	1.2	—	—	—	—	—	—	83	78	73	68	62	57	54	50
7.6	1.5	—	—	—	—	—	—	92	85	79	74	69	64	59	55
9.1	1.7	—	—	—	—	—	—	99	93	87	80	74	69	64	59
10.7	1.8	—	—	—	—	118	111	106	98	92	85	79	73	68	64
12.2	1.8	—	—	139	131	125	118	112	104	97	90	84	78	—	—
13.7	2.0	159	151	145	137	131	123	117	109	102	94	88	82	—	—
15.2	2.1	165	159	151	144	130	130	122	115	107	99	92	—	—	—
16.8	2.3	172	165	158	150	135	135	126	118	111	103	—	—	—	—
18.3	2.4	178	170	163	155	139	139	131	122	115	107	—	—	—	—
19.8	2.6	184	177	168	160	144	144	135	126	117	109	—	—	—	—
21.3	2.6	189	182	173	165	148	148	139	130	121	113	—	—	—	—
22.9	2.7	195	187	178	169	151	151	142	134	125	—	—	—	—	—
24.4	2.9	200	191	183	174	156	156	146	137	129	—	—	—	—	—
25.9	3.0	205	196	187	178	159	159	149	140	131	—	—	—	—	—
27.4	3.2	209	200	191	182	163	163	153	144	134	—	—	—	—	—
29.0	3.4	214	203	195	186	167	167	156	146	—	—	—	—	—	—
30.5	3.4	217	207	198	189	169	159	159	149	—	—	—	—	—	—
32.0	3.4	221	212	202	193	173	162	162	153	—	—	—	—	—	—

(Continued)

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Table E.10 (Concluded)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7	8
Minimum circumference at top, cm		**99**	**94**	**89**	**84**	**79**	**74**	**69**	**64**	**58**	**53**	**48**	**43**	**38**	**38**
Length of pole, m	**Groundline distance from butt, m ***	**Minimum circumference at 1.8 m from butt, cm**													
33.5	3.7	225	215	206	196	176	165	165	155	—	—	—	—	—	—
35.1	3.7	229	219	209	200	178	168	168	158	—	—	—	—	—	—
36.6	3.7	233	223	212	202	192	170	170	160	—	—	—	—	—	—
38.1	3.7	235	225	215	205	195	184	173	163	—	—	—	—	—	—

** These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

Notes:

1) *For marketing purposes, yellow cedar can be considered the same as western red cedar on a dimensional basis only, but such poles shall be identified by actual species.*

2) *See Note 2) to Table E.3 on yellow cedar.*

3) *Lodgepole pine poles will not normally include Classes H1 to H6, nor be taller than 15.2 m.*

4) *There is evidence to suggest that lodgepole pine poles of average strength lower than 46 MPa can be obtained from certain areas of the United States.*

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Table E.11
Dimensions of western hemlock and interior Douglas fir poles (fibre stress 47 MPa)
(See Clause E.1.)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7
Minimum circumference at top, in		99	94	89	84	79	74	69	64	58	53	48	43	38
Length of pole, m	Groundline distance from butt, m*	Minimum circumference at 1.8 m from butt, cm												
6.1	1.2	—	—	—	—	—	—	81	76	71	66	61	56	52
7.6	1.5	—	—	—	—	—	—	89	84	78	72	67	62	57
9.1	1.7	—	—	—	—	—	—	97	91	85	78	73	67	62
10.7	1.8	—	—	—	—	116	110	104	97	90	84	78	72	67
12.2	1.8	—	—	136	130	123	116	110	103	96	89	82	76	—
13.7	2.0	157	150	143	136	129	122	115	108	101	93	86	80	—
15.2	2.1	164	157	150	142	135	128	120	113	105	98	90	—	—
16.8	2.3	170	163	156	148	140	133	125	117	109	101	—	—	—
18.3	2.4	176	169	161	153	145	137	129	121	113	105	—	—	—
19.8	2.6	182	174	166	158	150	142	134	125	117	108	—	—	—
21.3	2.6	187	179	171	163	155	146	138	129	120	112	—	—	—
22.9	2.7	192	184	176	167	159	150	141	132	124	—	—	—	—
24.4	2.9	197	189	180	172	163	154	145	136	127	—	—	—	—
25.9	3.0	202	193	185	176	167	158	148	139	130	—	—	—	—
27.4	3.2	207	198	189	180	171	161	152	142	133	—	—	—	—
29.0	3.4	211	202	193	183	174	164	155	145	—	—	—	—	—
30.5	3.4	215	205	196	187	177	167	158	148	—	—	—	—	—

(Continued)

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Table E.11 (Concluded)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7
Minimum circumference at top, in		99	94	89	84	79	74	69	64	58	53	48	43	38
Length of pole, m	Groundline distance from butt, m*	Minimum circumference at 1.8 m from butt, cm												
32.0	3.4	219	210	200	191	181	171	161	151	—	—	—	—	—
33.5	3.7	223	213	203	194	184	174	163	153	—	—	—	—	—
35.1	3.7	226	216	207	197	187	176	166	156	—	—	—	—	—
36.6	3.7	230	220	210	200	189	179	169	158	—	—	—	—	—
38.1	3.7	233	223	213	203	192	182	171	160	—	—	—	—	—

* *These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

Note: *See Note 3) to Table E.3 for the classification of interior Douglas fir.*

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Table E.12
Dimensions of southern yellow and Scots pine poles (fibre stress 54 MPa)
(See Clause E.1.)

Class		1	2	3	4	5	6	7
Minimum circumference at top, cm		69	64	58	53	48	43	38
Length of pole, m	Groundline distance from butt, m*	Minimum circumference at 1.8 m from butt, cm						
6.1	1.2	78	73	68	63	58	54	50
7.6	1.5	86	80	75	69	64	59	55
9.1	1.7	93	87	81	75	69	64	59
10.7	1.8	99	93	86	80	74	69	64
12.2	1.8	105	98	91	85	79	73	67
13.7	2.0	110	103	96	89	82	76	71
15.2	2.1	115	107	100	93	86	80	74
16.8	2.3	119	112	104	97	90	—	—
18.3	2.4	123	116	108	100	—	—	—
19.8	2.6	127	119	111	103	—	—	—

** These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

Note: *For southern yellow pine, the average number of annual growth rings measured on the butt in the outer 5 cm of poles having a circumference of 95 cm or less at 1.8 m from the butt, and in the outer 7.5 cm of poles having a circumference of more than 90 cm at 1.8 m from the butt, shall be not less than 12 and 18 rings, respectively. However, pole butts with 8 to 10 and 12 to 15 rings, respectively, shall be acceptable if 50% or more latewood is present.*

As an alternative, these ring count and latewood measurements may be made on an increment core taken at 1.8 m from the butt directly above the place where the average number of annual rings is indicated on the butt surface.

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Table E.13
Dimensions of western larch and coastal Douglas fir poles (fibre stress 55 MPa)

(See Clause E.1.)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7
Minimum circumference at top, cm		**99**	**94**	**89**	**84**	**79**	**74**	**69**	**64**	**58**	**53**	**48**	**43**	**38**
Length of pole, m	**Groundline distance from butt, m***	**Minimum circumference at 1.8 m from butt, cm**												
6.1	1.2	—	—	—	—	—	—	79	74	69	64	59	54	50
7.6	1.5	—	—	—	—	—	—	85	80	75	70	65	59	55
9.1	1.7	—	—	—	—	—	—	93	87	82	75	70	64	60
10.7	1.8	—	—	—	—	111	106	99	93	87	80	74	69	64
12.2	1.8	—	—	130	123	117	111	104	98	92	85	79	73	—
13.7	2.0	149	148	136	130	123	116	109	103	96	89	83	76	—
15.2	2.1	155	149	141	135	129	121	115	107	99	93	87	—	—
16.8	2.3	162	154	148	140	132	126	118	111	103	97	—	—	—
18.3	2.4	167	159	151	145	137	130	122	115	107	99	—	—	—
19.8	2.6	172	164	156	149	141	134	126	118	111	103	—	—	—
21.3	2.6	176	169	162	154	145	137	130	122	115	106	—	—	—
22.9	2.7	181	173	165	158	150	141	134	125	117	—	—	—	—
24.4	2.9	184	177	169	162	153	145	137	129	120	—	—	—	—
25.9	3.0	189	182	173	165	156	149	140	131	122	—	—	—	—
27.4	3.2	193	186	177	169	160	151	142	135	125	—	—	—	—
29.0	3.4	197	189	181	172	164	155	145	137	—	—	—	—	—
30.5	3.4	201	193	184	176	167	158	149	140	—	—	—	—	—
32.0	3.4	205	196	188	179	170	160	151	142	—	—	—	—	—

(Continued)

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Table E.13 (Concluded)

Class		H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7
Minimum circumference at top, cm		**99**	**94**	**89**	**84**	**79**	**74**	**69**	**64**	**58**	**53**	**48**	**43**	**38**
Length of pole, m	**Groundline distance from butt, m***	**Minimum circumference at 1.8 m from butt, cm**												
33.5	3.7	209	200	191	182	173	164	154	145	—	—	—	—	—
35.1	3.7	212	203	195	184	176	167	156	148	—	—	—	—	—
36.6	3.7	216	206	197	188	178	169	159	150	—	—	—	—	—
38.1	3.7	219	210	200	191	181	172	162	151	—	—	—	—	—

* *These distances are for use only when a definition of groundline is necessary to apply requirements relating to scars, straightness, etc.*

Note: *See Note 3) to Table E.3 for the classification of coastal Douglas fir.*

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Table E.14
Dimensions of eastern white cedar reinforcing stubs (fibre stress 26 MPa)
(See Clause E.1.)

Length of stub, m	Minimum circumference at 1.8 m from butt, cm			
	Class Z	Class Y	Class X	Class W
3.0	113	98	84	71
3.4	129	112	96	81
4.0	143	124	106	—
4.7	149	129	—	—

Table E.15
Dimensions of spruce and eastern white pine reinforcing stubs (fibre stress 34 MPa)
(See Clause E.1.)

Length of stub, m	Minimum circumference at 1.8 m from butt, cm			
	Class Z	Class Y	Class X	Class W
3.0	103	89	76	64
3.4	117	102	87	74
4.0	129	112	96	—
4.7	135	117	100	—

Table E.16
Dimensions of amabilis fir reinforcing stubs (fibre stress 37 MPa)
(See Clause E.1.)

Length of stub, m	Minimum circumference at 1.8 m from butt, cm			
	Class Z	Class Y	Class X	Class W
3.0	99	86	73	62
3.4	113	98	84	—
4.0	125	109	94	—
4.7	130	113	—	—
5.9	144	125	—	—

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Table E.17
Dimensions of western red cedar and ponderosa pine reinforcing stubs (fibre stress 39 MPa)
(See Clause E.1.)

Length of stub, m	Minimum circumference at 1.8 m from butt, cm			
	Class Z	Class Y	Class X	Class W
3.0	98	85	72	61
3.4	112	97	83	—
4.0	123	107	91	—
4.7	128	112	—	—
5.9	142	123	—	—

Table E.18
Dimensions of red pine reinforcing stubs (fibre stress 41 MPa)
(See Clause E.1.)

Length of stub, m	Minimum circumference at 1.8 m from butt, cm			
	Class Z	Class Y	Class X	Class W
3.0	96	83	71	60
3.4	109	95	81	69
4.0	120	105	89	76
4.7	125	109	93	—

Table E.19
Dimensions of Jack pine reinforcing stubs (fibre stress 44 MPa)
(See Clause E.1.)

Length of stub, m	Minimum circumference at 1.8 m from butt, cm			
	Class Z	Class Y	Class X	Class W
3.0	94	81	69	58
3.4	107	94	79	—
4.0	118	102	87	—
4.7	123	106	—	—

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Table E.20
Dimensions of yellow cedar and lodgepole pine reinforcing stubs (fibre stress 46 MPa)

(See Clause E.1.)

Length of stub, m	Minimum circumference at 1.8 m from butt, cm			
	Class Z	Class Y	Class X	Class W
3.0	93	80	68	58
3.4	106	92	78	—
4.0	116	101	86	—
4.7	121	105	—	—
5.9	134	116	—	—

Table E.21
Dimensions of western hemlock and interior Douglas fir reinforcing stubs (fibre stress 47 MPa)

(See Clause E.1.)

Length of stub, m	Minimum circumference at 1.8 m from butt, cm			
	Class Z	Class Y	Class X	Class W
3.0	92	80	68	57
3.4	107	91	77	—
4.0	115	100	85	—
4.7	120	104	—	—
5.9	133	115	—	—

Table E.22
Dimensions of southern yellow and Scots pine reinforcing stubs (fibre stress 54 MPa)

(See Clause E.1.)

Length of stub, m	Minimum circumference at 1.8 m from butt, cm			
	Class Z	Class Y	Class X	Class W
3.0	88	76	64	54
3.4	100	86	74	64
4.0	110	95	81	69
4.7	114	99	85	—

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Table E.23
Dimensions of western larch and coastal Douglas fir reinforcing stubs (fibre stress 55 MPa)

(See Clause E.1.)

Length of stub, m	Minimum circumference at 1.8 m from butt, cm			
	Class Z	Class Y	Class X	Class W
3.0	87	75	64	54
3.4	99	86	73	—
4.0	109	94	80	—
4.7	113	98	—	—
5.9	125	109	—	—

Note: *See Note 3) to Table E.3 for the classification of coastal Douglas fir.*

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